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RESEARCH PAPER • SMALL-MODEL REASONING

AGOPD: Advantage-gated online distillation—mitigating teacher-supervised negative transfer in small-model mathematical reasoning

AGOPD: **Advantage-Gated On-Policy Distillation** for Mitigating Negative Transfer in Small-Model Mathematical Reasoning

AGOPD-RL Research Group
Independent empirical study • working paper

Student Qwen3-1.7B Base **Teacher** Qwen3-4B-GRPO **Data** DAPO-Math-17K
Main Eval 1,024 prompts • paired

ABSTRACT

On-policy distillation (OPD) can provide dense token-level teacher supervision on self-generated trajectories of student models, but the value of this supervision depends on the student state faced by the teacher. This paper takes Qwen3-1.7B Base and DAPO-Math without post-training as the study subject to examine the negative transfer phenomenon when OPD is jointly trained with outcome-reward-based reinforcement learning. Under a unified non-thinking decoding protocol, \(n=8\) sampled 16,164 training questions one by one, and estimated the question-by-question ability based on the number of correct answers in eight samplings; 48.9% of the questions did not get the correct answer in eight samplings (95% CI [48.1%, 49.7%]), and the success rate of the remaining questions spanned the entire 0/8–8/8 interval. Under the condition of sharing training data, training budget and evaluation protocol, the accuracy of GRPO baseline on fixed 1,024 questions is 28.42%; after unconditionally adding OPD, it drops to 24.12%, which is 4.30 percentage points lower than GRPO (paired bootstrap 95% CI [−6.93, −1.66]). Correct chat-template retests showed that fixed teachers maintained a positive item-level outcome advantage across five student ability ranges; in contrast, the state intervention experiment showed that teacher rescue rates decreased as the error prefix deepened, and the length matching control further showed that the impairment mainly came from incorrect reasoning semantics rather than context length. Based on the above distinction, this paper proposes **Advantage-Gated On-Policy Distillation** (AGOPD): only enable candidate distillation for student trajectories with negative group relative advantages, and require the teacher model to independently solve the same problem and pass the same result verifier. The accuracy of AGOPD reached 27.05%, which was 2.93 percentage points higher than that of ungated RL+OPD (95% CI [+0.29, +5.57]); its remaining difference with GRPO did not reach statistical significance in the current pairing evaluation. The above results indicate that the scope of teacher supervision must consider both outcome-level capabilities and state-level recoverability.

Keywords: on-policy distillation; reinforcement learning; GRPO; mathematical reasoning; conditional teacher advantage; negative transfer; small language model

48.9% Standardized n=8 0/8 successful training questions in the census	−4.30pp Unconditional RL+OPD significantly decreases relative to GRPO
+2.93pp Significant recovery of AGOPD relative to RL+OPD	40.2% The average trajectory rows activated by negative dominant gates in main training

1

Introduction

Rewards for verifiable results have become an important source of supervision for post-training mathematical reasoning. Methods represented by GRPO and DAPO construct policy optimization signals without relying on independent value models by sampling the same problem multiple times and making relative comparisons within the group [1] [2]. Online policy distillation (OPD) introduces teacher-distribution supervision on the states actually visited by the student model to alleviate the state distribution shift in offline distillation [4]. Two types of goals provide **outcome-level supervision** and **token-level supervision** respectively: the result reward gives a verifiable correctness signal for the entire answer, and the token-level teacher distribution gives a conditional distribution target for each prefix position. The former is sparse but reliable, and can be infinitely expanded with tasks; the latter is dense, but provided by a fixed teacher model, and its correctness does not automatically establish with the student status. Therefore, the benefits of training the two together need to be verified by controlled experiments rather than a priori assumptions.

However, the effectiveness of joint training relies on an implicit premise - the teacher model has teaching value in the state of student-visited. To test this premise, this paper conducts controlled experiments on Qwen3-1.7B Base and DAPO-Math-17K without post-training: the four methods share training subsets, optimized configurations and fixed evaluation protocols, and differ only in whether teacher supervision is superimposed. The results show that unconditionally combining OPD with result-supervised reinforcement learning (RL+OPD) results in significant performance degradation: GRPO's accuracy on the fixed 1,024 verification questions is 28.42%, RL+OPD is only 24.12%, and question-by-question pairwise statistics give −4.30 percentage points (95% bootstrap confidence interval [−6.93, −1.66], McNemar's exact test p=0.0018). In other words, in a state where students have been able to develop effective strategy updates, unconditional teacher supervision may interfere with rather than promote learning.

Negative transfer cannot be explained by teachers' item-level abilities alone. After recalibration, teachers maintain a stable advantage in different students' ability ranges when answering from the root state of the question; however, when teachers continue to deal with students' reasoning states that have deviated from the correct path, their recoverability continues to decline as the wrong path deepens. Length-matching controls further suggest that this degradation is primarily related to the semantic state carried by the erroneous inference, rather than the context lengthening itself. As a result, this paper shifts the focus of analysis from "whether teachers are stronger" to "in what state is teacher supervision still reliable?"

Based on the above distinction, this paper proposes **Advantage-Gated On-Policy Distillation** (AGOPD). Its design directly reuses the group relative advantages calculated during the joint reinforcement learning process: only negative advantage student trajectories are considered as distillation candidates; then the fixed teacher model is required to independently solve the same problem and pass the result validator consistent with the student model. Only when both the student-side negative advantage condition and the teacher-side verifiable correctness condition are met, the teacher's token-level target is imposed on the corresponding student-visited state. This design does not change the underlying GRPO advantage estimator or train additional routing models, but instead limits the non-zero scope of the distillation target to the intersection of outcome-level verifiability and trajectory-level relative disadvantage.

The main contributions of this paper are as follows:

- **Controlled negative transfer measurement.** In the fixed 1,024-item paired assessment, the unconditional superposition of teacher **token-level supervision** brought measurable performance degradation on an item-by-item basis, turning negative transfer from an empirical phenomenon into a testable statistical conclusion (see Table 4 for pairing statistics).
- **Question-level and state-level diagnosis.** Ability stratification under the correct protocol shows that fixed teachers maintain a non-negative outcome advantage in each interval; the counterfactual error prefix experiment further shows that question-level teacher ability cannot guarantee state-level rescue reliability.
- **Advantage gated distillation.** AGOPD combines group relative advantage and verifiable teacher correctness into two-level routing conditions, brings measurable recovery based on ungated RL+OPD, and clearly limits its scope of application and statistical boundaries.
- **Mechanism verification at the learning level.** In a comparison of strict alignment between the training budget and the evaluation protocol, as measured by semantic accuracy (the final answer is mathematically equivalent to the reference answer, it is counted as correct, and the output format is not critical). The sequence-level supervision of learning a complete and verifiable correct trajectory from the question root state is better than imposing the direction of the teacher's token distribution on the student's access prefix; the statistical uncertainty still covers zero difference, and this result is only used as a learning layer support for the state misalignment explanation, rather than an independent significance conclusion.

Accordingly, this paper puts forward four testable research propositions (Propositions 1–4): (1) Unconditional superposition of OPD on a stable GRPO baseline will cause measurable performance degradation on a topic-by-question basis; (2) Question-level teacher advantages cannot fully predict the reliability of supervision on student-visited state; (3) Double gating consisting of group relative advantage and verifiable teacher correctness can alleviate the above-mentioned negative transfer; (4) AGOPD's function is to limit the scope of the distillation target, and its applicability is still restricted by the student training stage and the teacher's ability boundary. Chapters 3 and 5 will organize the evidence according to these claims.

2

Tasks, Data, and Basic Methods

2.1 Task and Data Construction

This paper studies outcome-supervised reinforcement learning for mathematical reasoning: given the competition mathematics question x, the student model needs to generate an answer ending in the line $\texttt{Answer:}$, which is judged right or wrong by the rule validator. The training questions are from DAPO-Math-17K, with a total of 16,164 questions after cleaning; the fixed validation set of 1,024 tracks is used for comparison of all methods (see § 5.1 and Table 3 for the protocol).

Data objects are divided into three categories based on their purpose. First, 16,164 cleaned training questions were used in the standardized ability census to estimate the question-by-question success rate $\hat p(x)$ of basic students and diagnose the signal density of outcome supervision within the group; its discrete distribution is shown in Table A1. Second, the main four-arm experiment used a fixed training pool of 1,472 questions, and the four methods shared the training pool, optimized configuration, and random seeds, thus limiting the method differences to teacher supervision and its routing methods. Third, the difficult bucket experiment uses 1,120 $\hat p=0.125$ questions and 2,000 randomly selected $\hat p=0$ questions, a total of 3,120 questions, to test the training boundary of the low student success rate area. The 1,472-item training pool was derived from the early frontier screening process and was not a direct threshold slice of the standardized census; this source difference and its impact are described separately in Appendix A.

2.2 Group-Relative Reinforcement Learning under Outcome Supervision

Given a math problem x, the current student policy π_θ samples G responses $y_i\sim\pi_\theta(\cdot\mid x)$ to the same problem. As a result the validator returns the reward $r_i=V(x, y_i)$. The main experiment of this paper uses binary correctness rewards and calculates the standardized relative advantage within the sampling group of the same problem:

$$A_i=\frac{r_i-\mu_r}{\sigma_r+\epsilon},\qquad \mu_r=\frac{1}{G}\sum_{j=1}^{G}r_j \tag{1}$$

A_i in equation (1) describes the performance of the i trajectory relative to other sampled trajectories of the same problem, rather than the absolute difficulty of the problem. This intra-group relativization is the design basis for GRPO to omit explicit critic: multiple sampling of the same question is used as the baseline, which avoids the storage and stability overhead of training the value network on each state. For binary rewards, positive and negative contrasts are formed only when success and failure occur simultaneously within the group; if the rewards within the group are completely consistent, the normalized group relative signal will degrade.

2.3 Effective Group Rate and Training-Signal Density

Assume that the single success probability of a certain question under the current strategy is p. For an independent sampling group of size G, the probability that the group contains at least one successful trajectory and one failed trajectory is defined as the Effective Group Rate (EGR):

$$\operatorname{EGR}(p, G)=1-(1-p)^G-p^G \tag{2}$$

It can be seen from equation (2) that when $p\rightarrow0$ or $p\rightarrow1$, EGR approaches 0. Therefore, the data set average accuracy does not directly reflect the effective signal density of group relative reinforcement learning. This paper only uses EGR as a training signal diagnostic quantity and does not use it as a gating rule for AGOPD.

2.4 On-Policy Distillation

Online policy distillation computes the teacher distribution on trajectories generated by the student model itself. Let the state of the student track i before the token t be $s_{i, t}=(x, y_{i,\lt t})$. Its general form is the token-level student–teacher distribution divergence:

$$\ell_{\mathrm{OPD}}(i, t)=D\bigl(\pi_\theta(\cdot\mid s_{i, t})\,\|\,\pi_T(\cdot\mid s_{i, t})\bigr) \tag{3}$$

divergence D. Two common examples are reverse KL and hard target cross entropy; the hard target takes the maximum probability token $z^T_{i, t}=\arg\max_v\pi_T(v\mid s_{i, t})$ distributed by the teacher, and takes $-\log\pi_\theta(z^T_{i, t}\mid s_{i, t})$ as the target (the main experiment of this paper uses hard targets, see §4.3). This training method enables teacher supervision to cover the states actually visited by the student policy, but at the same time introduces a premise: the local conditional distribution provided by the teacher model on these states should have teaching value [4] - §3 will directly test this premise.

2.5 Conditional Teacher Advantage

In order to characterize the relative effectiveness of the teacher model, this paper defines conditional teacher advantages on the student ability interval b:

$$M_T(b)=\operatorname{Acc}_T(b)-\operatorname{Acc}_S(b) \tag{4}$$

When $M_T(b)>0$, the average outcome accuracy of the teacher model in this ability range is higher than that of the student model. It should be noted that the difference in the accuracy of $M_T(b)$ measurement results is not equivalent to the token-level distillation quality; even with $M_T(b)>0$, the teacher's conditional distribution on the student's error prefixes may still not be able to effectively correct the current status. Therefore, this paper regards the item-level ability advantage as a necessary but insufficient diagnostic quantity, and tests item-level ability and state-level recoverability in §3.2 and Appendix F respectively.

3 Mechanistic Diagnosis of Negative Transfer under Unconditional RL+OPD

In order to establish negative transfer as the object of mechanism discussion, this section first reports the phenomenon of the controlled four-arm experiment (see Table 4 for values, see §5.1 for protocol and statistical metric definition): under shared data, training budget and evaluation protocol, the result is that supervised RL has a stable gain compared to the untrained Base; unconditionally superimposing teacher **token-level supervision** (RL+OPD) on it shows a significant decrease; the point estimate of pure OPD (without RL) does not exceed GRPO (see Table 4 for values and pairing statistics). This section then examines the mechanistic origins of this negative transfer. The core hypothesis is that basic students frequently enter the wrong reasoning state on the current mathematical distribution, and the token-level supervision of OPD happens to act on these student-visited states; even if the teacher has a higher outcome accuracy rate from the question root state, he may also lose part of the recoverability on the prefixes of students who have deviated from the correct path. In order to distinguish the three explanations of "teacher himself is not strong enough", "context lengthening" and "error state semantics", this paper reports in sequence the basic student ability census, the teacher ability comparison of the question root state, the error prefix depth experiment, the equal-length matching state control, and the controlled learning comparison of trajectory-level supervision from the question root state and online state-level supervision.

The mechanism assumes that incorrect exploration on the student side exposes online distillation to states that deviate from the correct reasoning path; if the teacher's conditional recoverability on these states decreases, unconditional

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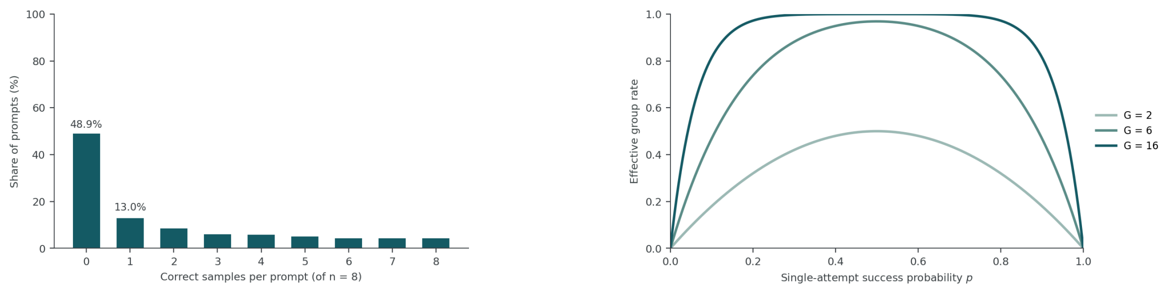
may become a source of negative transfer. This hypothesis needs to be supported by both state-level counterfactual evidence and controlled experiments at the learning level, and cannot be derived solely from the overall accuracy of teachers or students.

3.1 Ability distribution and result signals of basic students

For the 16,164 cleaned training questions, this paper uses Qwen3-1.7B Base (denoted as S_0) without post-training to sample each question $n=8$ times under the unified non-thinking decoding protocol. The temperature is set to 0.6, and $\hat p(x)=k/8$ is used to estimate the success rate of each question. The questions with all failures (0/8) account for the highest proportion, and the success rate distribution covers the entire 0/8–8/8 interval. The complete discrete distribution is shown in Table A1 and Figure 1(a). Combining the effective group rate of equation (2) (Figure 1(b)), it can be seen that for questions with concentrated failures, simply increasing the number of optimization steps does not always form an effective positive and negative group comparison; only questions with a medium-to-high success rate have stable intra-group signals. This result shows that: the outcome-level signal is sparse on a considerable part of the questions, but the questions with above-average success rate still account for a considerable proportion; the wrong trajectory is still a common state that needs to be faced in training.



(a) Capability census of the base model (n = 8)

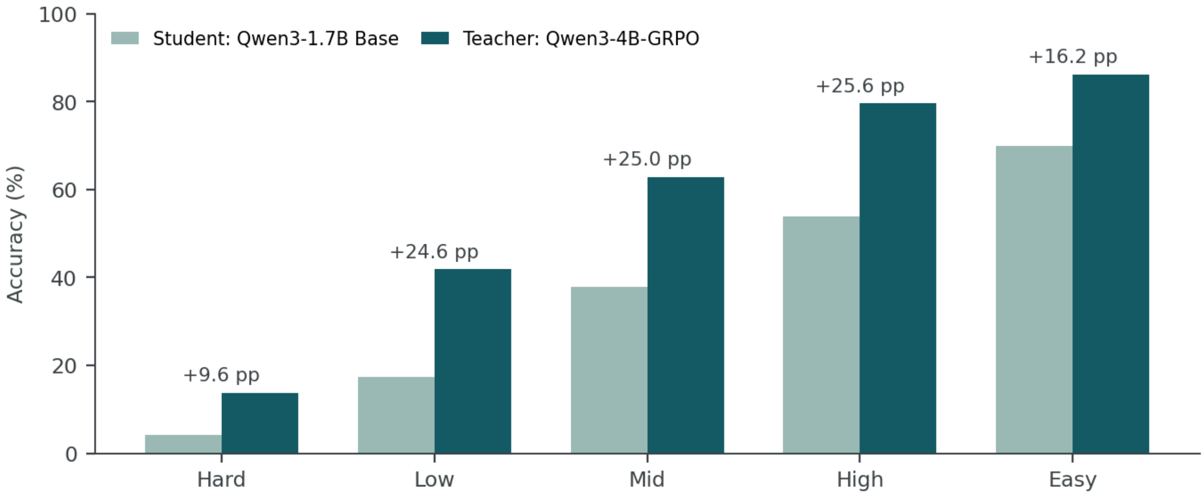
(b) Effective group rate

Figure 1 • Ability distribution and effective group rate.(a) Discrete success number distribution of $n=8$ on 16,164 training questions of Qwen3-1.7B Base under the standardized non-thinking protocol.(b) The theoretical effective group rate corresponding to different sampling group sizes G under the binary result reward. The concentration of capability quality in the 0/8 region means that simply increasing the number of training steps does not guarantee a positive and negative contrast for each sampling group.

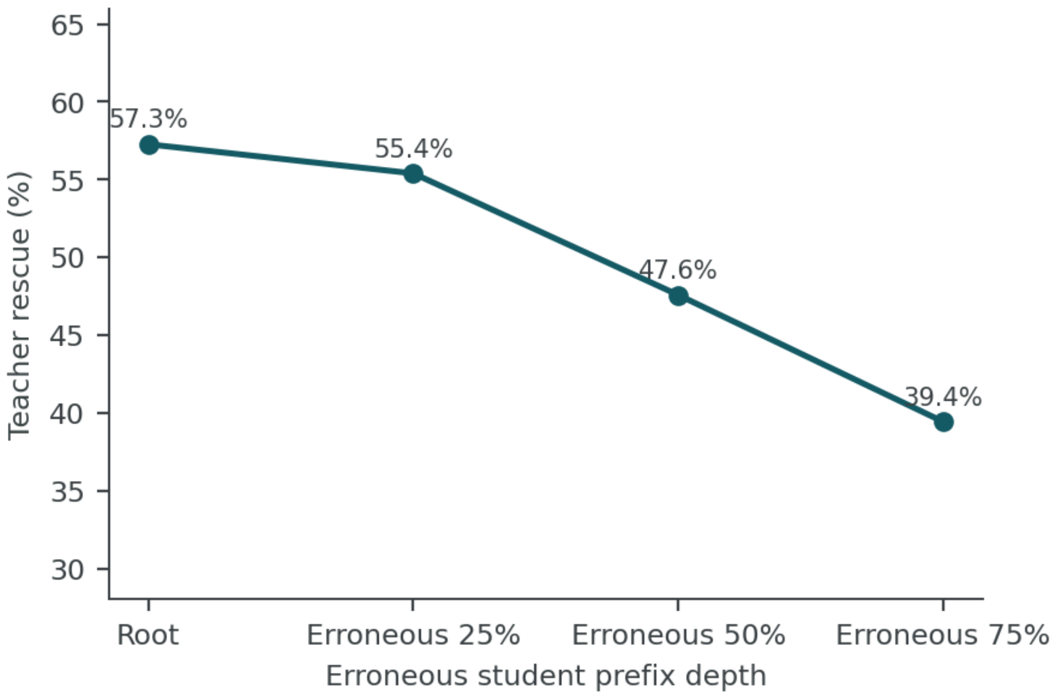
This census is only used to characterize the student-side ability structure and the availability of group relative learning signals; the fixed training pool of the main four-arm experiment has an independent source of historical screening. The data relationship between the two is shown in Appendix A.

3.2 Stable Teacher Advantage at the Question Root State

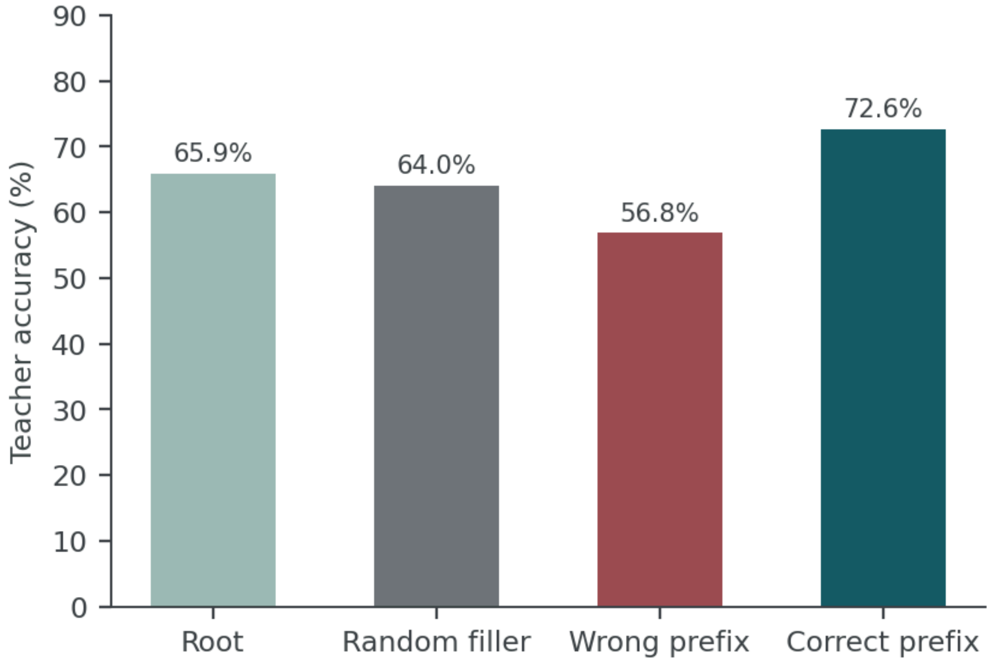
If negative transfer mainly comes from teachers' absolute lack of ability, then teachers should also have obvious ability gaps in the root status of the questions. To test this explanation, this paper retests the same questions on five intervals divided by Base students' abilities, and keeps the decoding agreement between students and teachers consistent.



(a) Teacher advantage at the problem root



(b) Teacher rescue vs. erroneous-prefix depth



(c) Length-matched state control

Figure 2 • Teacher supervision changes from item-level ability to state-level recoverability.(a) Teachers maintain a question-level advantage in the five Base ability intervals; (b) As the incorrect student prefixes deepen, the teacher's rescue rate continues to decrease; (c) Length matching control shows that the damage of incorrect prefixes is significantly greater than that of equal-length random contexts, while the correct prefixes have a promoting effect. Panels (b) and (c) use different sample sets, so the two Root baselines are not directly compared numerically.

The results show that when teachers answer questions from the root state, they maintain a positive result advantage in all ability ranges, and this advantage does not disappear as students' ability increases - it still maintains a range close to the average of the entire sample in the two higher ability ranges of Mid and High (see Figure 2(a) for breakdown of each range). Therefore, the failure of unconditional superposition OPD cannot simply be attributed to teachers "not knowing how to do these questions"; what really needs to be answered is whether this question-level advantage can be extended to an intermediate state where students have deviated from the correct path.

3.3 Accumulated Loss of Teacher Recoverability on Erroneous Student States

Question-level advantages are not sufficient to ensure reliable state-level supervision. In order to directly observe the teacher's performance after being brought to the error state by the students, this paper intercepts the prefixes based on the proportion of the total length of the student's error path (25%, 50%, 75%) as the teacher's input, and compares it with the root state of the question. The teacher is required to continue writing from there and judge it according to the same result verifier. The teacher rescue rate continues to decrease as the error prefix deepens, and the decrease does not only appear suddenly at a certain depth, but gradually accumulates (Figure 2(b)), forming a damage path that intensifies with depth.

The above decline can still be explained by a simpler mechanism: longer prefixes themselves increase the burden of inference. In order to separate "context length" and "state semantics", this paper selects questions that have both correct and incorrect prefixes, and the two types of prefixes are of equal length, and constructs three sets of comparisons: incorrect prefixes, random text padding of the same length, and correct prefixes. Random padding and wrong prefixes only differ in whether they carry wrong inference semantics, and the length effect can be measured separately; correct prefixes are used to test the semantic direction.

Under equal-length matching, the incorrect prefix makes the teacher's accuracy significantly lower than that of the random text, while the correct prefix makes the teacher's accuracy significantly higher than the incorrect prefix, and there is no discernible difference between the random text and the question root state (Figure 2(c)). If the damage is caused only by the lengthening of the prefix, the wrong prefix should be equivalent to random text; the actual difference shows that what determines the teacher's recoverability is the inference semantics carried by the prefix - wrong semantics weakens subsequent rescue, and correct semantics provides a favorable intermediate state.

> Two sets of state-level evidence experiments play complementary roles: depth sweep gives cumulative damage evidence that "the deeper the error, the lower the rescue rate"; the equal-length matching control further illustrates that the damage comes from incorrect inference semantics rather than context length. Together, they attribute the impairment to faulty inference of semantics rather than context length.

3.4 Directional verification at the learning level: trajectory-level supervision and online state-level supervision

The above state evidence further leads to a learning-level prediction: if part of the damage of unconditional online state-level supervision (on-policy distillation, OPD) comes from the misalignment of supervision on the wrong state, then trajectory-level supervision (rejection-sampling fine-tuning, RFT) from the root state of the question, with the complete verifiable correct trajectory as the supervision object, should be able to reduce this exposure. In order to test this prediction, this paper constructs a controlled comparison between RFT and OPD: the two share Qwen3-1.7B Base initialization, the same 120-question training pool, batch size 8, 120 optimization steps, learning rate 1e−6, maximum sequence length 4,096 and seed 42. The only difference is the source and granularity of the trajectory on which supervision relies - RFT learns the complete correct trajectory starting from the root state of the question, OPDApply the teacher's token distribution on the student's prefix. The assessment uses a fixed set of 1,024 questions with the same decoding protocol.

The results are reported in terms of semantic accuracy: this metric definition counts answers that are mathematically equivalent to the reference answer but do not strictly meet the established output format as correct. Therefore, it measures whether students can give correct conclusions without requiring them to strictly conform to a certain format contract. Statistics were conducted using paired bootstrap 95% intervals and McNemar's exact test for each question.

Table 2 • Controlled learning comparison between trajectory-level supervision (RFT) and online state-level supervision (OPD) (semantic accuracy, fixed 1,024 questions). The test was for question-by-question pairwise differences.

Method	Semantic verifier	Δ vs OPD	95% CI	McNemar p
OPD (state-level, online)	265 / 1,024 (25.88%)	—	—	—
RFT (root, trajectory-level)	292 / 1,024 (28.52%)	+2.64 pp	[−0.10, +5.37]	0.0664

As measured by semantic accuracy, trajectory-level supervision (RFT) is 2.64 percentage points higher than online state-level supervision (OPD), and the direction is consistent with state misalignment prediction; however, the paired interval still covers zero difference (McNemar p=0.066). The current single seed, 120-step evidence is not enough to claim that RFT is significantly better, and can only support the directional conclusion that "learning complete trajectories from the root state may reduce the exposure of error states."

> Conclusions in this section: Existing evidence positions negative transfer as a state-dependent supervision problem: the teacher has a capability advantage in the root state of the question, but this advantage does not automatically extend to the state where the student has deviated from the correct path; the deeper the error state, the weaker the teacher's recoverability, and isometric control shows that the main damage comes from incorrect reasoning semantics rather than context length. Controlled results from trajectory-level supervision provide additional, yet significant, evidence in the same direction. Therefore, this paper regards state misalignment as one of the direct mechanisms of OPD negative migration, rather than the only cause. This distinction between the question level and the state level corresponds to the introduction claim 2; the learning layer comparison in § 3.4 shows the same direction but not yet significant support, see Table 2.

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4 AGOPD: Advantage-Gated On-Policy Distillation

AGOPD does not change the basic distillation divergence, but changes teacher supervision from global application to conditional application: whether it is applied is determined by the relative advantages of the student side group and the verifiable correctness of the teacher side. The overall process is shown in Figure 3. See \(\S4.1\)–\(\S4.3\) for specific definitions.

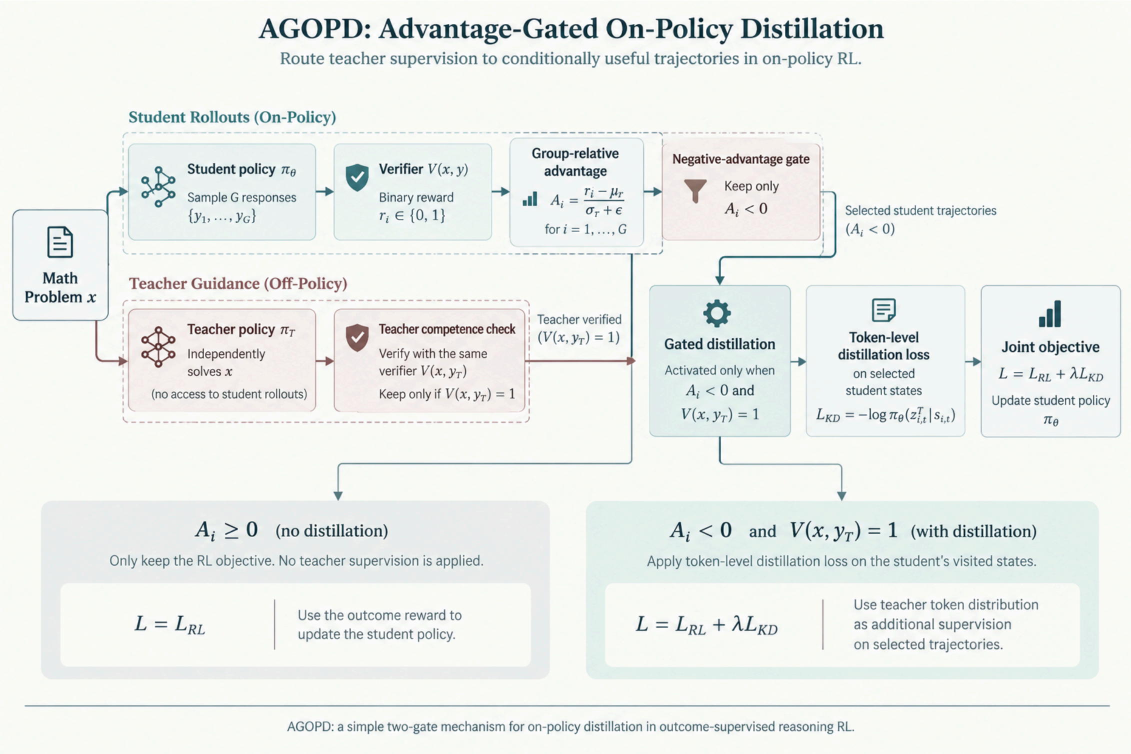


Figure 3 • AGOPD method structure. The negative advantage gate selects the trajectory of relatively disadvantaged students at the trajectory level; the teacher competency gate requires the teacher model to solve independently and pass the same result verifier at the question level. Only student-visited states that pass both levels of gating receive non-zero distillation weights.

4.1 Negative-Advantage Trajectory Gating

For the i student sampling trajectory, define the negative advantage gate:

$$g_A(i)=\mathbb I[A_i<\tau_A] \tag{5}$$

The main experiment uses $\tau_A=0$. Therefore, a trajectory enters the candidate distillation set only if its group relative advantage is negative. In the implementation, the negative advantage amplitude is further used to construct the trajectory weight:

$$w_A(i)=\min\bigl(\max(-A_i,0), w_{\max}\bigr) \tag{6}$$

Among them $w_{\max}=2.0$. Within the same problem, the trajectory that achieves the relative advantage of the non-negative group is only optimized by the reinforcement learning objective; the distillation weight of the negative advantage trajectory increases with its relative disadvantage and is truncated by $w_{\max}$.

It is necessary to clarify the function boundary of the gating: when a group of samples all fails, the binary reward group is exactly the same, the standardized advantage $A_i=0$, the negative advantage gate of equation (5) is not activated, and distillation will not occur. This type of "no signal within the group" problem belongs to the sampling and data level (signal density of $\S3.1$), and is not the scope of application of AGOPD; AGOPD deals with relatively disadvantaged trajectories in mixed result groups.

4.2 Verifiable Teacher-Competence Gating

Negative advantage only indicates that the student trajectory performed worse relative to the same sample and does not guarantee that the fixed teacher model has higher reliability on this problem. Therefore, for questions containing candidate trajectories, the teacher model π_T first independently generates the complete answer:

$$ y_T\sim\pi_T(\cdot\mid x) \tag{7} $$

Then use a validator consistent with the student model result reward to determine whether the teacher's answer is correct:

$$ g_T(x)=\mathbb I[V(x, y_T)=1] \tag{8} $$

The final distillation activation variable is defined as:

$$ g_i=g_A(i)\, g_T(x) \tag{9} $$

If the independent solution of the teacher model fails the result validator, the distillation loss of all candidate trajectories under this problem is set to zero. What this gating ensures is that the teacher model can independently obtain verifiable correct answers at the question level, while the the teacher's token distribution is locally optimal on every student prefix.

4.3 Distillation objective and joint optimization

Assume that the state of the i student trajectory before the token position t is $s_{i, t}=(x, y_{i,\lt t})$. For the student-visited state that passes the two-level gating, the main experiment uses the token with the highest probability of the teacher model as the hard target:

$$ \begin{gathered} z^T_{i, t}=\arg\max_v\pi_T(v\mid s_{i, t})\\ \ell_{\mathrm{KD}}(i, t)=-\log\pi_\theta(z^T_{i, t}\mid s_{i, t}) \end{gathered} \tag{10} $$

The teacher's goal is calculated on the prefixes actually visited by the student model, thus retaining the state coverage property of on-policy distillation; whether to apply this **token-level supervision** is determined by the two-level gating of Equation (5) and Equation (8).

$$ \mathcal L_{\mathrm{KD}}= \frac{\sum_i g_i w_A(i)\sum_t m_{i, t}\ell_{\mathrm{KD}}(i, t)} {\sum_i g_i w_A(i)\sum_t m_{i, t}+\epsilon} \tag{11} $$

Among them, $m_{i, t}$ is the response token mask. The final objective function adds the gated distillation term to the original GRPO objective:

$$\boxed{\mathcal L_{\mathrm{AGOPD}}=\mathcal L_{\mathrm{GRPO}}+\lambda_D\mathcal L_{\mathrm{KD}}} \tag{12}$$

$\lambda_D=1.0$ in the main experiment. Therefore, AGOPD differs from standard hard-target cross-entropy distillation not in the form of the loss, but in limiting the non-zero scope of the distillation target to the set of trajectories that satisfy:

$$\mathcal S_{\mathrm{AGOPD}}=\{(x, y_i): A_i\lt0\ \land\ V(x, y_T)=1\} \tag{13}$$

The hard target of Equation (10) is applied position by position according to the entire context of the selected trajectory: at the token position at which the student is correct, the teacher argmax is consistent with the token that the student has generated. This loss term degenerates into self-distillation anchoring - it does not inject new signals, but anchors the strategy near itself, keeping the policy entropy within the observable range without losing control. This property is what distinguishes AGOPD from variants such as "distilling only at divergent positions" or switching to soft targets. The evidence of its training dynamics is shown in Appendix I.

Algorithm 1 · Advantage-Gated On-Policy Distillation

Input: student policy π_θ, teacher π_T, verifier V, prompt batch $\mathcal B$, group size G, threshold $\tau_A=0$.

1. For each question $x\in\mathcal B$, G responses are sampled online by the student model.
2. The validator calculates the binary result reward, and obtains the relative advantage A_i within the group according to formula (1).
3. Standard GRPO policy goals are calculated for all student trajectories.
4. Select the trajectory of $A_i<\tau_A$ according to equation (5), and weight it according to $w_A(i)$ in equation (6).
5. If a problem contains at least one candidate trajectory, the teacher model independently solves the problem.
6. Check the teacher's answer with the validator of equation (8); if it fails, the distillation corresponding to the problem will be closed.
7. When the verification passes, the token-level distillation loss of Equation (10) is calculated on the prefix state of the candidate trajectory.
8. Jointly update the student model parameters according to Equation (12).

4.4 Scope of Applicability

The scope of application of AGOPD requires a clear distinction between two types of problems. First, it reduces the chance of continuing to impose teacher supervision on trajectories where the student model has achieved non-negative group relative advantage; second, it filters out problems that the teacher model cannot correctly solve independently. AGOPD does not modify the reward estimator of GRPO, nor does it construct additional positive samples for the all-failure group. When the binary rewards are exactly the same within the group, the relative advantage of the standardized group is 0, and the negative advantage gate of equation (5) will not be activated. Therefore, ability frontier screening, sampling group size and stable reinforcement learning configuration solve "whether there is a comparable signal within the group"; AGOPD solves which student trajectories teacher supervision should act on after relative advantage has been generated.

5

Experiments

This section organizes the experimental evidence around the four research propositions proposed in the introduction, and reviews them one by one in each subsection and at the end of the chapter.

5.1 Experimental setup

Model. The trainable students of the main experiment are initialized from Qwen3-1.7B Base S_0 without post-training; pure GRPO, pure OPD, RL+OPD and AGOPD only differ in teacher supervision and routing methods. Fixed the 50-step checkpoint T of the teacher Qwen3-4B-GRPO. Both the main experiment and the main assessment were conducted in non-thinking mode. This paper does not interpret this teacher model as being a universally superior teacher to the larger base model; teacher ordering is not consistent across different validation subsets, so its role is limited to the within-task fixed teacher used in this study.

data. The training pool comes from DAPO-Math-17K and contains 16,164 questions after cleaning. The main four-arm experiment shares the same learnable subset of 1,472 questions (see § 2.1 and Appendix A for construction metric definition).

Training configuration. The main control uses full-parameter FSDP update, the learning rate is 10^-6, the KL loss coefficient is 0.05, the training batch size is 8, the sampling group size of each question is $G=6$, and the upper limit of the response length is 2,048.

The sampling parameters are temperature 0.6, top-p 0.95 and top-k 20. The main results and gating statistics both use the same 180 optimization step training trajectory; the GRPO baseline checkpoint referred to in this paper refers to the 180th step checkpoint. For the sake of unification, this paper records the pure GRPO main line as 180-step GRPO, and only uses this metric definition in the text, tables and illustrations.

Evaluation and statistics. The main validation set contains a fixed 1,024 questions, and all methods share seed 42, temperature 0.6, top-p 0.95, top-k 20, and an output upper limit of 2,048 tokens. Since the comparison is based on the same batch of questions, this paper uses question-by-question paired bootstrap (100,000 resamplings) supplemented by McNemar's exact test. The current experiment only included a single training random seed, so these intervals characterize question-by-question uncertainty on a fixed training run rather than training randomness.

Table 3 • Main experimental protocol. For complete operating parameters, see the configuration file and startup script in the supplementary materials.

Item	Main setting	Notes
Student	Qwen3-1.7B Base	full-parameter update from S_0
Teacher	Qwen3-4B-GRPO-50step	fixed teacher
Train subset	1,472 prompts	all methods share the same historical frontier-oriented subset
Batch / group	8 / 6	48 student trajectories per optimization step
Optimizer	AdamW, lr=1e-6	KL loss coefficient 0.05
Decode	T=0.6, top-p=.95, top-k=20	max response 2,048
Main checkpoint	step 180	logs continue to 180 steps
Main evaluation	1,024 fixed prompts	paired statistics

5.2 Main Results: Gating Mitigates Negative Transfer from Unconditional Distillation

To keep the notation consistent with the method name, the “GRPO” row in Table 4 is denoted S_{RL}: it is the post-training reference initialized from the common S_0 using only outcome reward reinforcement learning under the same training pool and optimization protocol. The remaining four arms all start from the same S_0, and the only difference comes from whether to add teacher supervision and its routing method.

Table 4 • Fixed 1,024 subject results. The question-by-question paired confidence interval and McNemar's exact test both use GRPO as the main reference; the difference between AGOPD and GRPO does not reach statistical significance, and this result is not equivalent to statistical equivalence.

Method	Correct / 1024	Accuracy	Δ vs GRPO	Paired 95% CI	McNemar p
Base	239	23.34%	**−5.08pp**	[−7.71, −2.44]	0.00025
GRPO	291	**28.42%**	—	—	—
OPD	263	25.68%	−2.73pp	[−5.47, 0.00]	0.0622
RL + OPD	247	**24.12%**	**−4.30pp**	**[−6.93, −1.66]**	**0.00184**
AGOPD	277	**27.05%**	−1.37pp	[−4.10, +1.37]	0.360

The main comparison in Table 4 gives two sets of core conclusions. First, the resulting supervised RL has a stable gain compared to the untrained Base, but unconditionally superimposing teacher **token-level supervision** (RL+OPD) on top of it will significantly lower the accuracy (the pairing interval does not include 0). Second, double gating (AGOPD) recovers most of this loss, but does not surpass pure outcome supervision: the confidence interval of the remaining difference relative to GRPO still spans 0, so it cannot be judged that the distilled target is better than pure RL. The fixed assessment and question-by-question pairing statistics are shown in Table 4.

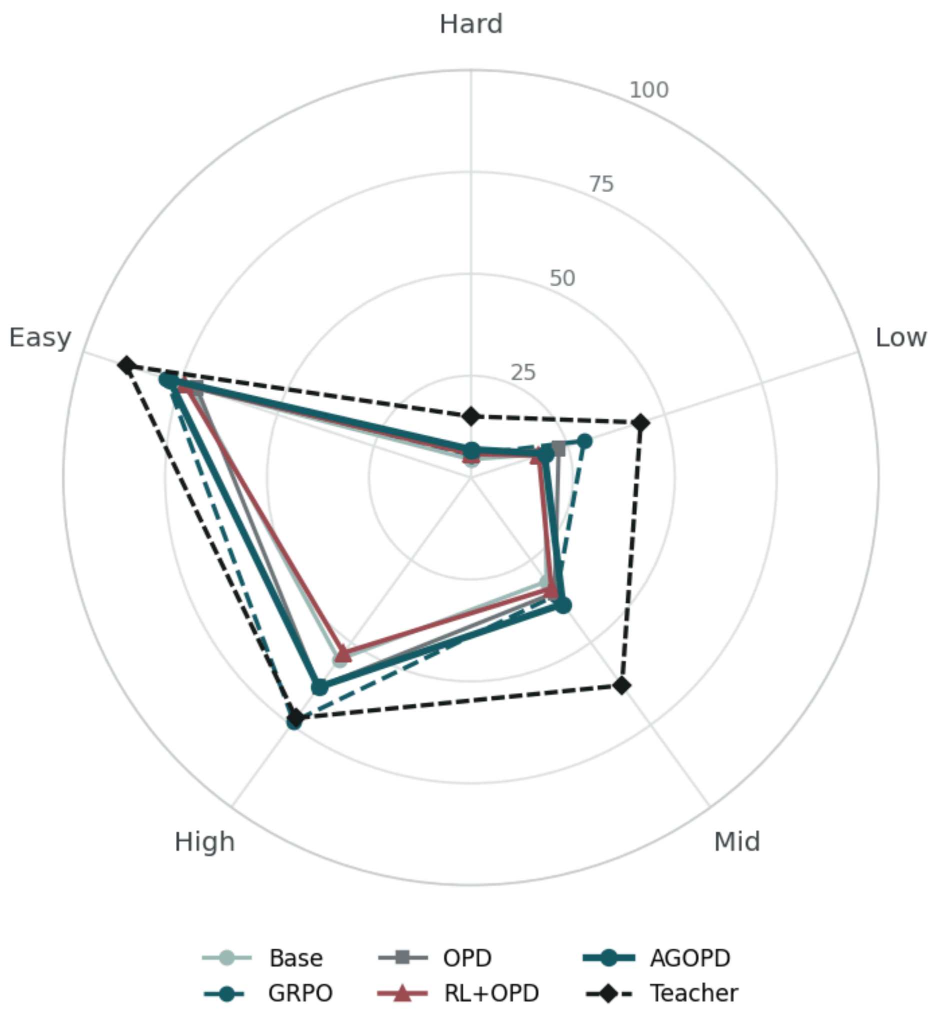


Figure 4 • Difficulty-tiered ability profile (radar). The five radii correspond to the five intervals (Hard→Easy) divided by the Base student \(n=8\) ability estimate. Each polyline gives the strict verification accuracy of a model within this interval; the teacher has a fixed Qwen3-4B-GRPO-50 checkpoint, which is the same fixed 1,024 question metric definition as the student's five methods. The teacher profile is located at the outermost level in most intervals, and only the High interval is slightly lower than the GRPO.

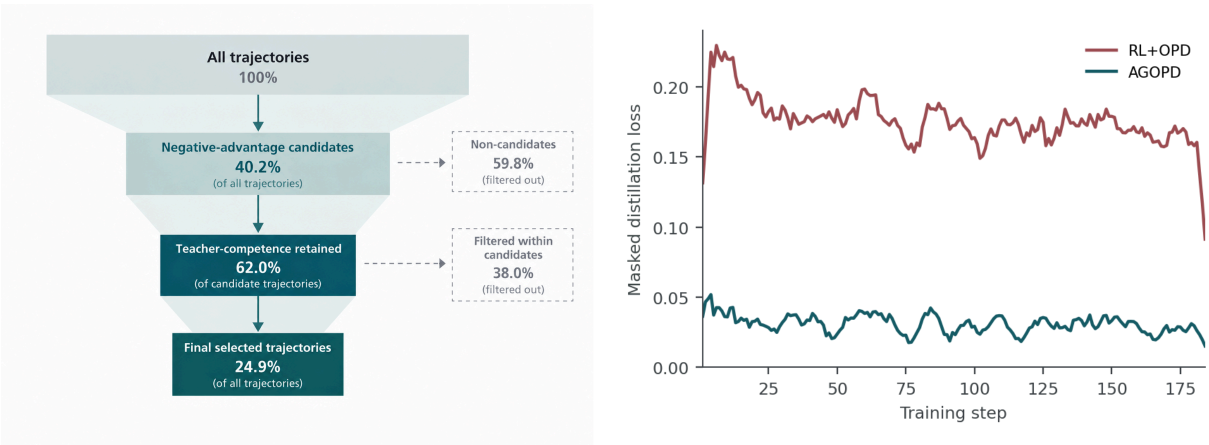
The difficulty layered profile in Figure 4 shows that the recovery of AGOPD is non-uniform: its point estimates in Hard and Mid are higher than the corresponding GRPO, close to GRPO in Easy, significantly lower in Low, and only partially recover the loss of RL+OPD in High; the fixed teacher profile is located in the outermost layer in most intervals. No interval-level significance analysis has been performed. This profile is only used as a descriptive capability structure. This paper does not claim any stable advantages on intervals; see Appendix G for the robustness of multiple sampling.

MAIN RESULTS

The empirical role of AGOPD is to alleviate the negative migration introduced by unconditional RL+OPD, rather than to prove that distilled targets are generally better than pure GRPO.

5.3 Gating dynamics and distillation scope

The recovery difference of § 5.2 requires a mechanistic explanation. The statistics of the whole training process (Figure 5) show that the negative advantage gate releases about 40% of the trajectories into candidates on average, and the teacher's competent gate retains about 62%, which reduces the average distillation loss of AGOPD to about one-fifth of that of unconditional RL+OPD (0.177→0.032), and the policy entropy of the two is almost the same (0.1577 vs. 0.1571).



(a) Gate selectivity

(b) Distillation loss

Figure 5 • Gating dynamics and distillation scope.(a) Report the proportion of negative dominant candidates after aggregation of 180 training steps, and the proportion of teachers who are qualified and retained based on the candidate set.(b) Seven-step moving average of the distillation loss recorded in the training log; the distillation loss is affected by the mask, trajectory weight and sample difficulty at the same time, and is only used as a training diagnostic quantity and not as an independent estimate of the gradient scope.

These diagnoses are consistent with the "scope narrowing" explanation: less supervision and higher fixed validation results (Table 4, +2.93pp), so the removed supervision (imposed on trajectories where the advantage is positive or teacher-unsolvable) is more likely to be net harmful; the entropies are nearly identical ruling out the alternative explanation that the gain comes from global entropy suppression. It should be noted that the above relationship belongs to process diagnosis; causal proof at the mechanism level still requires the ablation of independent components of two-level gating (see §7.5). The conclusion that AGOPD is superior to RL+OPD is based on the paired test in Table 4.

5.4 Gated Distillation on Difficulty Buckets: Bounds Testing in Regions of Low Student Success Rates

The results in §5.2 and §5.3 suggest that the benefits of teacher gating need to be examined separately in areas of low student success. In order to measure this boundary, this paper constructs a difficult bucket training set: take all 1,120 questions of $\hat p=0.125$ in the ability census, and randomly select 2,000 questions of $\hat p=0$ (seed 42), a total of 3,120 questions. This construct was chosen because the low-success area has both a weaker student outcome signal and a lower teacher absolute correct rate; it is used for boundary diagnosis and is not to be interpreted as a direct test of a change in the sign of teacher relative advantage.

Using the soft target variant of AGOPD of §4 (this variant replaces the hard target of §4.3 with a temperature-sharpened teacher soft distribution and is only used for this boundary test), continue training on this bucket for 180 steps from the first round of RL checkpoints. The strict verification evaluation accuracy of 1,024 questions and 2,048 token budgets is 28.22% (95% of the paired intervals contain zero points relative to the GRPO baseline), which is 28.42% from the starting point. There is no significant improvement or significant damage. Combined with the negative transfer conclusion in §5.2, this result shows that the role of gated distillation is to avoid damage caused by unconditional superposition, rather than to create proven additional gains in areas where the teacher's absolute accuracy is low; the feasibility of trajectory-level transfer is shown in Appendix D and Appendix F. The relationship between the two scopes on the difficulty axis is shown in Figure 6 (schematic, not data).

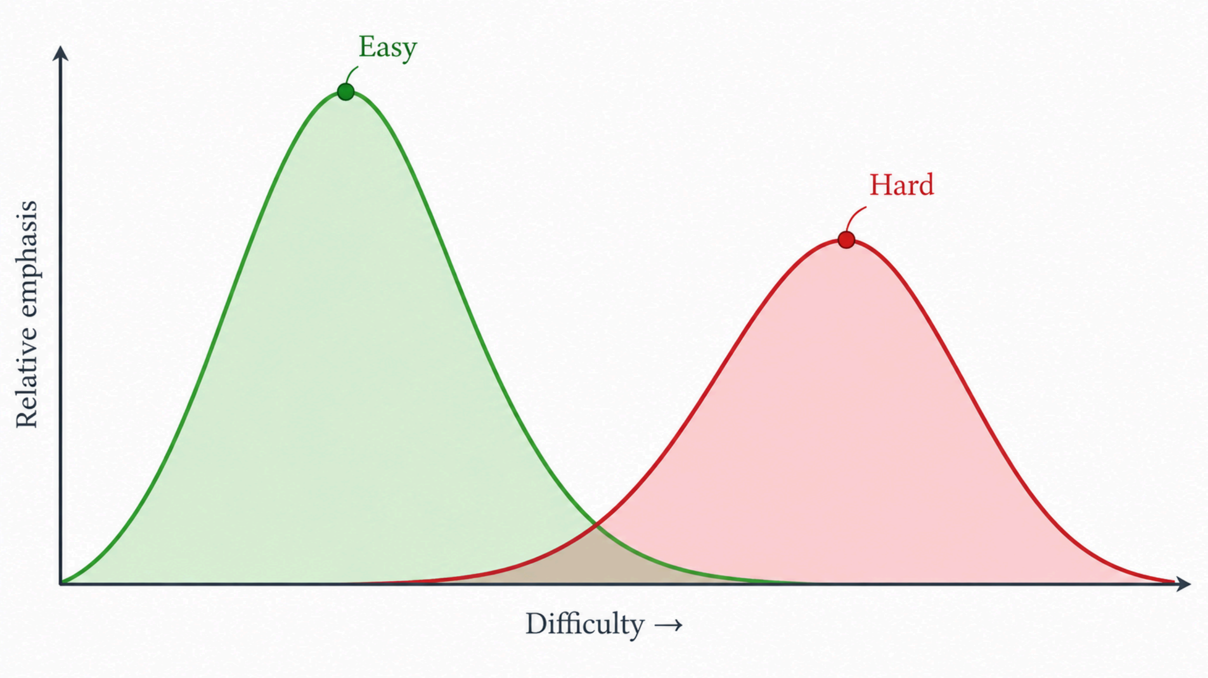


Figure 6 • Schematic representation of gated distillation scope (non-data graph). The horizontal axis increases in difficulty to the right, and the vertical axis indicates emphasis; red is the area of high difficulty and low student success rate (difficulty bucket, gated difficulty has additional gains), and green is the easier area (comfort pool). This figure is only a conceptual representation and does not correspond to any specific values.

> Evidence-claim comparison. The study claims are organized according to the introduction: the topic-by-question evidence for Claim 1 is in the RL+OPD row of Table 4 (relative GRPO −4.30pp, the pairing interval does not include 0); Claim 2 is undertaken by Chapter 3 (question-level teacher advantages are shown in Figure 2(a), and state-level impairment is shown in Figure 2(b)/(c)); Claim 3 is in the AGOPD row of Table 4 (+2.93pp recovery) and the non-uniform recovery profile of Figure 4; Claim 4 is covered by § 5.3The gating statistics and distillation loss (scope terms) are assumed with the difficult bucket boundaries of § 5.4, where the dimensions of the student training phase are shown in Appendix D.

6

Related Work

6.1 Outcome-Supervised Reasoning Reinforcement Learning

An important advantage of reinforcement learning for mathematical reasoning is that the final answer can be directly determined by rules or automatic verifiers. DeepSeekMath proposed GRPO to replace the independent value model with the relative advantage within the group, thereby reducing the training overhead of reasoning reinforcement learning[1]. On this basis, DAPO further studies stability issues such as dynamic sampling, clipping strategies, token-level strategy gradients, and ultra-long response control [2]. DeepSeek-R1 demonstrates the scalability of verifiable rewards in large-scale long-chain inference training [3]. Different from the above work, this paper does not modify the advantage estimation mechanism of GRPO, but studies which student trajectories the teacher signal should act on when token-level teacher supervision is combined with result-supervised reinforcement learning.

6.2 Online policy distillation and reinforcement learning-distillation joint training

The Generalized Knowledge Distillation (GKD) proposed by Agarwal et al. calculates teacher feedback on the self-generated output of the student model to alleviate the offset between the training distribution and inference distribution of the autoregressive model, and discusses its joint use with reinforcement learning fine-tuning [4]. Subsequently, G-OPD interprets on-policy distillation as a special case of dense reinforcement learning with KL constraints and extends the standard OPD [5] with reward scaling and reference model design. This type of method mainly adjusts the form or relative weight of the distillation target; AGOPD retains the basic GRPO and distillation target, focusing on changing the non-zero scope of the distillation term.

6.3 Capability-aware and capability frontier distillation

PACED clearly models the question-level success rate of the student model, and points out that the quality of the distillation gradient will decline in the too-hard and too-easy areas, so it uses the ability frontier weighting based on the success rate to make the training more focused on the student's proximal development area [7]. SEAD uses teacher-student entropy difference and curriculum mechanism to implement ability-aware supervision [8] from three scales: token, training stage and question. These works illustrate that "distillation needs to be conditioned according to students' abilities" is not a view unique to this paper. The difference of AGOPD is that it directly reuses the calculated trajectory-by-trajectory group relative advantage in the joint reinforcement learning batch as the student-side selection signal, and uses the same result validator for question-level teacher competency checking. Therefore, AGOPD's gating variables come from the relative outcome signal of the current reinforcement learning trajectory, rather than offline success rate estimates or entropy statistics.

6.4 Teacher uncertainty, token distribution and online distillation dynamics

EOPD dynamically adjusts forward and reverse KL for high teacher entropy words to improve the generation diversity [6]; Tail-Aware OPD points out that only renormalizing the top-k probabilities will lose tail distribution information, and explicitly restores the tail probability quality [10]. Rethinking OPD analyzes the effective conditions for online distillation [9] from the perspectives of teacher-student reasoning mode compatibility, whether the teacher has students' missing abilities, and the coverage of student-visited states. Its follow-up work further discusses the state coverage and alignment speed [11] under the condition of very few training samples. These works focus on the quality of teacher distribution at the word level or state level; the conditional teacher advantage and result verification gate in this paper are located at the question level and trajectory level, and the two types of signals are complementary. Question-level teacher correctness does not guarantee that the teacher provides the optimal word distribution for each wrong student prefix, which is also an important limitation of AGOPD.

7

Discussion and Limitations

7.1 Insufficient intra-group comparison and teacher routing are two different bottlenecks

The all-failure group does not activate the negative advantage gate, which is a data and sampling bottleneck of "whether there is a learnable contrast within the group" (see §3.1); which trajectories that have produced relative advantages should be targeted by teacher **token-level supervision** is a method scope issue. The two have been clearly distinguished in §4.4, and this limitation will not be expanded upon.

7.2 Verifiable teacher correctness is only a proxy variable for the teacher’s local quality

The teacher competency gate requires the teacher model to independently solve the problem from scratch and pass the result validator. This condition can filter out cases where the teacher model cannot correctly complete the entire problem, but it does not prove that the teacher provides a low-noise, locally consistent conditional distribution on any student prefix \(s_{i, t}\). The student's trajectory may have entered an unrecoverable error branch, and the teacher's solution from the beginning is correct does not mean that its word distribution under the error prefix condition is still the optimal teaching signal. More fine-grained routing can further incorporate teacher entropy, student-teacher state compatibility, prefix recoverability, or local verification signals.

7.3 Teacher Effectiveness Relative to a Continuously Changing Student Policy

Stage-2 multi-stage experiments show that effective fixed teachers in the first stage do not guarantee that the same gains will continue to occur at subsequent student checkpoints. This phenomenon is consistent with the basic idea of conditional teacher advantage: teacher quality is not an absolute attribute but is defined relative to current student strategies, training distributions, and decoding protocols. The Stage-2 results of this paper are not sufficient to attribute changes in performance solely to teacher obsolescence, but they clearly illustrate that teacher selection and competency judgments need to be re-evaluated as the training stages progress.

7.4 Data sources and decoding modes limit the scope of extrapolation

The 1,472-item historical training subset used in the main four-arm experiment was formed before the standardized ability census, and its early screening artifacts did not uniformly and explicitly turn off thinking mode at all stages. Therefore, the 16,164-item non-thinking ability census should be viewed as a diagnostic of signal structure rather than a strict generative rule for the main training subset. In addition, there is a significant difference in absolute ability between non-thinking and thinking modes, so the total failure proportion conclusions of this paper only apply to the standardized non-thinking protocol.

7.5 Statistical Scope and Missing Ablations

The current main results come from a single training random seed. Question-by-question paired bootstrap and McNemar tests can quantify the uncertainty of fixed training checkpoints on validation questions, but are not a substitute for independent training repetitions. Formal large-scale validation should prioritize increasing independent training seeds for GRPO, RL+OPD, and AGOPD and supplementing component ablation using only negative advantage gates or only teacher-competent gates to distinguish the respective contributions of the two levels of gating.

7.6 Computational overhead

AGOPD needs to call the fixed teacher model on the negative dominant candidate problem and calculate the distillation target on the student-visited state, so its computational overhead is higher than that of pure GRPO. Teacher competency gates reduce ineffective teacher supervision but still introduce additional reasoning and token-level forward computations. Follow-up work is needed to evaluate the compute-performance trade-offs between gating thresholds, teacher call ratios, and capacity recovery.

8

Conclusion

This paper studies the scope of conditions applicable to teacher supervision in outcome-supervised reinforcement learning. In the controlled experiment of Qwen3-1.7B Base initialization without post-training, the relatively stable GRPO of unconditional RL+OPD showed significant question-by-question performance degradation (Table 4); the fixed teacher maintained a non-negative advantage in question-level results, but this advantage was significantly compromised in the student error state. Based on this distinction, AGOPD limits the distillation scope with two-level gating of group relative advantage and verifiable teacher correctness, bringing measurable recovery based on ungated RL+OPD, while the remaining difference relative to pure GRPO does not reach significance in the current pairing evaluation.

The evidence can be summarized as a causal chain: the negative transfer of unconditional online distillation is significant and measurable; gating can eliminate this damage, but exposes the upper limit of the ability of fixed teacher token-level distillation in this setting - AGOPD does not exceed pure outcome supervision; the multi-stage and trajectory-level exploration in Appendix D further suggests that injecting complete trajectories at the sequence level under long output budgets may bring about positive transfer, but this part is an exploratory point estimate (non-paired, incomparable across budgets), and no final conclusion is made.

Ability Census and Data Sources

A.1 Census protocol and discrete capacity distribution

This paper uses a standardized non-thinking protocol to sample $n=8$ from the 16,164 cleaned training questions one by one, and defines $\hat p(x)=k/8$ by the number of times it passes the result verifier in eight samplings. Table A1 gives the complete discrete distribution. This census was used for signal density supervision of diagnostic outcomes and was not used as additional training data for the main four-arm method comparison.

Table A1 • Complete discrete distribution of the standardized 16,164-item ability census. The official statistical metric definition is based on the nine discrete results of this table. In the Unified Ability Census, there are 6,162 questions that satisfy $\hat p\ge0.25$, so the 1,472 question main training subset is not interpreted as a direct slice of this threshold.

Success count / 8	Prompts	Fraction
0/8	7,905	48.90%
1/8	2,097	12.97%
2/8	1,383	8.56%
3/8	985	6.09%
4/8	928	5.74%
5/8	809	5.00%
6/8	694	4.29%
7/8	684	4.23%
8/8	679	4.20%

A.2 Training pool sources and experimental boundaries

The training subset of the main four-arm experiment consisted of 1,472 questions from the frontier screening process prior to the standardized census. It is a fixed training pool shared by the four methods, rather than a set directly sliced by the current census threshold. For the difficult bucket experiment, all 1,120 $\hat p=0.125$ questions were taken, and 2,000 questions (seed 42) were randomly selected from the $\hat p=0$ area, for a total of 3,120 questions. Recording the three types of data objects separately avoids confusing diagnostic census, main training pool, and boundary experiments into the same data generation process.

Therefore, the conclusion in the text about all failed questions only refers to the census distribution of the 0/8 bin in Table A1; the causal comparison of the main four-arm experiment comes from the shared 1,472-question training pool; the difficult bucket results only answer the boundary question of the low student success rate area. The agreement relationship, sample purpose and comparable scope of the three should not replace each other.

APPENDIX B

Optimization-Configuration Selection and Stability Diagnosis

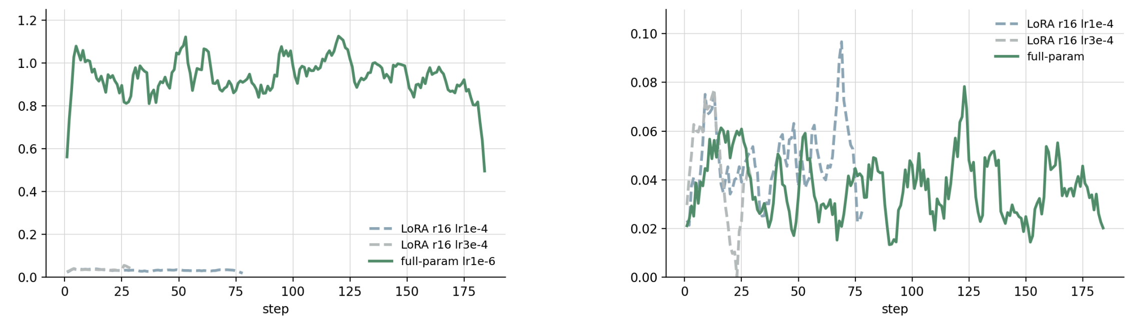
This appendix explains how the optimal configuration for the primary four-arm experiment is determined. Diagnostic experiments vary the update channel, learning rate, and KL constraints while keeping the task, model size, and basic sampling protocol unchanged; these results are used to exclude obviously unstable training paths and are not used as evidence of performance of AGOPD. Table B1 describes the observations, usage, and final disposition of each configuration to avoid misinterpreting exploratory failed runs as method results.

B.1 Debugging chain: from update channel to learning rate and KL constraints

The process of determining the master agreement is carried out along the two lines of "trainable and movable → trained and stable". The first line is the update channel: the gradient norm of LoRA (r16) under lr=10 −4 and 3×10 −4 is only about 0.03, which is one order of magnitude lower than the 0.5–1.4 of the full parameters (Figure B1(a)), and the policy gradient loss is almost unchanged (Figure B1(b)). This type of configuration cannot produce an observable learning signal no matter how the LoRA learning rate is increased, so it is excluded in Table B1, and the main baseline shifts to full parameter update.

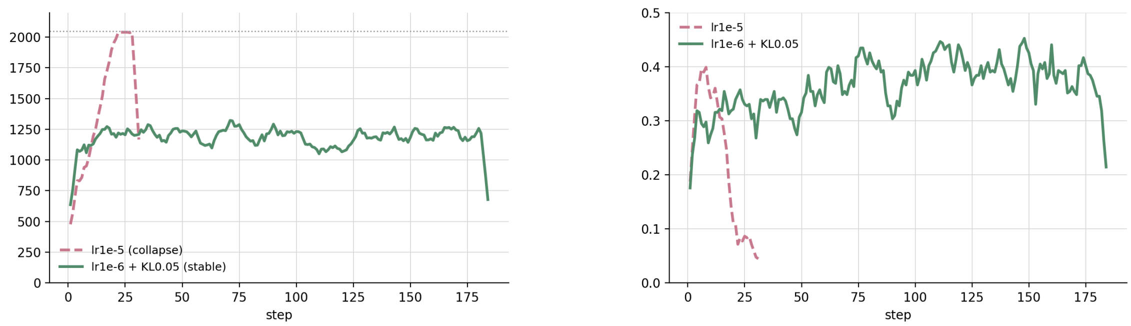
The second line is learning rate and regularization. Under full parameters, lr=3×10 −4 collapses in about 30 steps: the online score drops to zero, and the policy entropy first rises and then collapses (Figure B2(b)); lr=10 −5 is trainable in the early stage, but later enters a "drift-verbose-truncation" cycle, and the response length repeatedly hits the upper limit of 2048 (Figure B2(a)), and the score subsequently drops. The two failure modes point to "excessive updates" and "lack of constraints on policy drift" respectively, indicating that simply reducing the learning rate is not enough to ensure stability throughout the process.

Accordingly, the learning rate is reduced to 10 −6, and KL 0.05 is superimposed for anchoring: the policy entropy remains stable, the gradient is non-zero throughout, and the score rises monotonically (green solid line in Figure B1/B2), that is, it is frozen into a four-arm master protocol. The entire debugging line compresses "Why not use LoRA, why is it not a high learning rate, why is KL constraint needed" into a reproducible elimination process. In the end, only the frozen full parameter configuration is entered into the method comparison; the observation and treatment of each failed configuration are documented in Table B1.



(a) Gradient norm (b) Policy-gradient loss

Figure B1 • Update channel diagnostics.full-param lr=1e−6 maintains stable updates, LoRA (r16) effectively updates one order of magnitude weaker under lr=1e−4/3e−4: (a) gradient norm, (b) policy gradient loss.



(a) Response length (b) Online score

Figure B2 • Learning rate degradation and KL anchoring. The full parameter lr=1e−5 enters the drift-verbosity-truncation cycle and collapses, and lr=1e−6+KL 0.05 remains stable: (a) response length (including the upper limit of 2048), (b) online score.

Table B1 • Optimization configuration filtering. Candidate runs are used to determine a stable baseline; the main results only use the frozen full parameter configuration and do not use point estimates between candidate runs as method comparisons.

Configuration	Observed behavior	Use in the paper
LoRA, $r=16$, lr=10^{-4}	weak effective update; low gradient magnitude	rejected as the main full-parameter baseline
LoRA, $r=16$, lr=3×10^{-4}	increasing lr did not recover a reliable learning signal	rejected; retained as an optimization diagnostic
Full parameter, lr=10^{-6}, KL=0.05	stable entropy and nonzero policy gradient	frozen main protocol
Full parameter, higher lr	response-length drift and training degradation	rejected; not used for method comparison

APPENDIX C

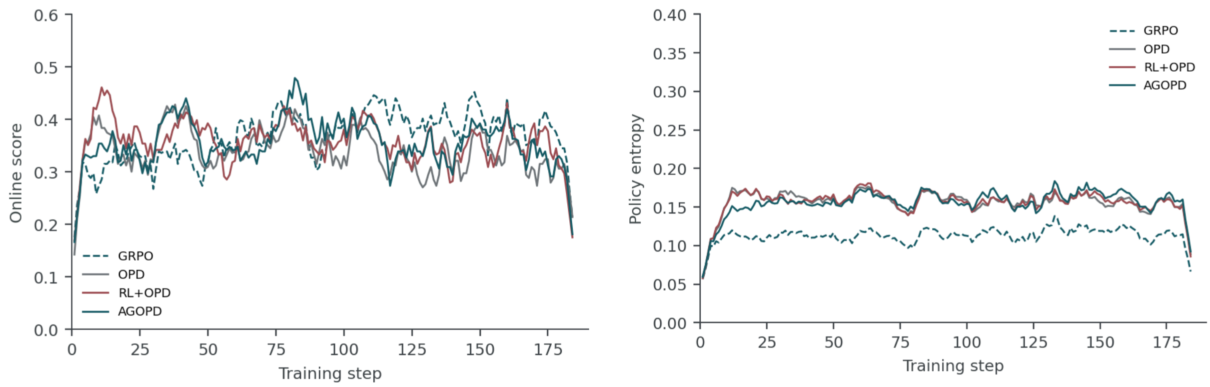
Main-Experiment Training Dynamics and Metric Definitions

This appendix uses training logs to test the health of the four-arm operation and defines the indicator metric definition used when referencing these logs in the text. Online score, policy entropy, response length and distillation loss fluctuate from optimization step to optimization step; here they are only regarded as running diagnostic quantities - comparisons between methods are always based on a fixed 1,024 question verification (Table 3/Table 4).

C.1 Four-method operational health check

None of the four main runs crashed or lost control of entropy during the entire training process (Figure C1). In terms of online scores, GRPO, RL+OPD and AGOPD all increased or remained high over time, and pure OPD (without RL) remained low and stable. In terms of policy entropy, the four methods are all in the healthy range. The entropy trajectories of AGOPD and RL+OPD are close (consistent with the explanation of "gating does not change global entropy" in §5.3 of the text), and the distillation-related volume has not increased abnormally. Stepwise recording of response lengths and truncation rates was used to confirm that there were no spurious increases due to budget pressure.

These observations provide operational-level endorsement for the method comparison in the main text: differences in the main results are not produced by collapse or truncation of individual runs.



(a) Online score

(b) Policy entropy

Figure C1 • Training health check for the four-line approach.(a) Online score, (b) policy entropy; the curve is a seven-step moving average, used to check whether there is a crash, entropy runaway or abnormal stage. Method comparison is still based on the fixed 1,024 question verification.

C.2 Indicator Dictionary

Table C1 gives the precise definition and diagnostic meaning of each indicator when these logs are quoted in the text.

Table C1 • Training dynamic indicator dictionary. Online indicators are used for operational diagnosis; only the verification results of 1,024 fixed questions are used for main conclusions.

Metric	Definition in the logs	Diagnostic interpretation
Online score	mean verified reward of sampled responses	batch-level training signal, not fixed-set accuracy
Policy entropy	token-level policy entropy on the response	detects concentration or entropy runaway
Response length / clip ratio	sampled response length and fraction at the cap	detects budget pressure and truncation
Distillation loss	masked teacher-student token loss	measures optimization pressure, not correctness

APPENDIX D

Trajectory-Level Transfer and Branch Comparisons under a Common Initialization

There is an essential difference between token-level online distillation and trajectory-level supervision in terms of supervision granularity and state conditions: the former imposes teacher token distribution on the prefixes actually visited by students, while the latter takes complete and verified trajectories as the object and updates students according to sequence-level goals in the question root state. The controlled comparison in §3.4 of the main text has provided directional evidence for trajectory-level supervision; this appendix tests whether this result holds true under a longer budget and two-stage setting, and clarifies the inheritance relationship between stages within the pedigree. To prevent the argument from being misled by its structure, this appendix separates "the construction of a common initialization" from "the controlled conditions that unfold from that initialization": the first two sections describe the former, and the last three sections report the latter.

D.1 Construction of trajectory-level supervision objects and initialization

The trajectory-level supervision object uses existing results to supervise students as a generation strategy. For each training question, the strategy independently samples n=8 times, and the result verifier determines whether each answer is correct or incorrect; the question-level acceptance rate for a question that can provide supervision samples is defined as the proportion of samples that pass the verification:

$$\hat\rho(x)=\frac{1}{n}\sum_{i=1}^{n}V(x, y_i),\qquad y_i\sim\pi_\theta(\cdot\mid x)\tag{D.1}$$

Using this strategy, all training questions are sampled and the verification passes are retained. Before cleaning to the standard answers, a set of sequence-level supervision samples carrying a complete problem-solving chain is obtained. The supervision object only contains trajectories that are ultimately correct and complete from the root state of the question, so students can learn the intermediate steps of the successful path rather than fitting local tokens on visited prefixes.

Repeated exposure of sequence-level supervision to a fixed distribution will cause overfitting, so the stopping time needs to be determined. The training was evaluated checkpoint by checkpoint with a fixed budget, and the accuracy showed a unimodal shape with the number of steps (Figure D1): the peak checkpoint was higher than the initial outcome supervision student's level under the same budget, and then fell back. The replay samples are highly concentrated in the low success rate area where students are almost completely wrong. In the later stage of training, repeated fitting on this narrow sub-distribution without new coverage constitutes sub-distribution over-fitting; the supervision signal itself does not change, what changes is only the repeated exposure of the model to the narrow sub-distribution. Peak checkpoints are selected accordingly as part of subsequent initialization.

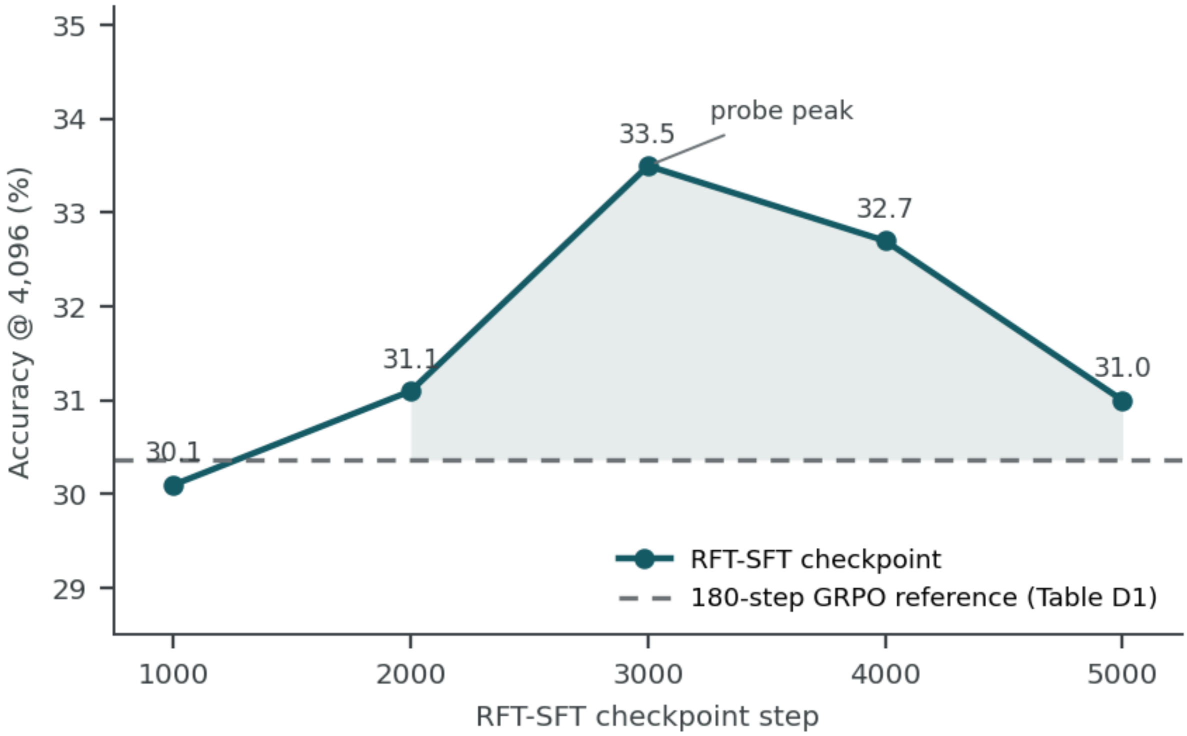


Figure D1 • RFT-SFT checkpoint probe (4,096 token budget, 1,024 fixed items). The solid line is the accuracy of the RFT-SFT checkpoint under the same budget, and the dotted line is the reference level of 180-step GRPO students (see Table D1 for values); the curve reaches a peak at about 3,000 steps and then falls back. The figure shows a checkpoint probe and is not to be interpreted as significant.

D.2 Common initialization and evaluation boundaries

The training checkpoints of the resulting supervised students and the peak checkpoints of the trajectory-level supervision are averaged by weight to form the common initialization of the two-stage training. It is necessary to clarify its evaluation boundaries: the averaged model is not re-evaluated, so the description of the common initialization can only refer to the probe value of the trajectory-level supervision peak checkpoint (the first row of Table D1), but cannot give the formal value of the initialization itself. This bound dictates that subsequent comparisons take the form of relative differences between branches rather than inferences about initialized absolute gains.

D.3 Branch comparison under common initialization

In order to test whether superimposing fixed-teacher online token distillation can produce additional benefits when trajectory-level hot start is already available, this paper starts from common initialization and compares three groups of controlled conditions under shared training data and budget: pure outcome supervision (GRPO), superimposed fixed-teacher gated distillation (AGOPD, the gating mechanism is the same as text § 4), and length-controlled GRPO that changes budget and length. The only differences between the three are limited to whether to use distillation and budget and length settings.

Table D1 ・ Common initialization and subsequent branch comparison. The first line is a merge initialization, and the subsequent three branches share this initialization. The only difference lies in whether to superimpose fixed teacher gated distillation (GRPO and AGOPD branches) and training budget and length settings (length-controlled branch). The @4,096 value in the first row is the RFT checkpoint probe rather than the merged re-evaluation, and @2,048 is not re-evaluation; each value is an exploratory point estimate under a fixed 1,024-question evaluation (single seed, non-paired, incomparable across budgets).

Stage / init & branch	Acc @2,048	Acc @4,096	Completion @2,048	Analysis role
Warm-start init (GRPO-180 ⊕ RFT-SFT ck3,000)	—	≈33.5 (ck3,000 probe)	—	common init (merged); post-merge eval not re-measured
Branch ・ GRPO (240 steps)	29.10%	33.20%	78.8%	2-phase result-RL from common init
Branch ・ AGOPD (240 steps)	27.05%	29.20%	69.6%	+ fixed-teacher gated distillation
Branch ・ GRPO, length-controlled	30.57%	31.35%	91.8%	budget/length sensitivity

The observations in Table D1 can be summarized into three points. First, the branch overlaid with fixed-teacher gated distillation is lower than the pure result-supervised branch under long budget, and also lower than the common initialization; second, the pure result-supervised branch and common initialization are roughly the same under long budget, showing no discernible absolute gain; third, the length-controlled branch improves the completion rate with short budget point estimation. The first point is consistent with the assertion that "the value of fixed teachers needs to be re-evaluated with the student stage", that is, when a strong hot start is already available, fixed teacher token distillation no longer provides effective gains. Limited by the evaluation boundary described in D.2, the above observations only support relative comparisons between branches and do not support any absolute gain expression.

D.4 Budget and length as independent dimensions

The length-controlled branch is used to separate the effects of budget, length and supervision shape. Starting from the same initialization, it trains under a longer output budget, and the completion rate increases significantly, while the long budget point estimate is not monotonically higher than the pure outcome supervision branch. This result suggests that length is a moderator variable independent of supervisory modality and does not alone account for ability gains.

D.5 Convergence and Unverified Comparisons

Based on the above observations: trajectory-level injection can exceed the results of uninjected supervision students under a long budget; when trajectory-level hot start is already available, continuing to superimpose fixed teacher online word distillation will not produce gain. The two observations jointly support the conclusion direction of § 8 of the main text about "the form of supervision determines returns, and fixed teachers need to be re-evaluated with the stages."Constrained by assessment boundaries, there are three items in this appendix that are not directly verified: co-initialized formal reassessments, teacher changes in relative student ability in Phase 2, and item-by-item pairwise statistics across budgets. Therefore, the above observations are directional point estimates and do not constitute a significant conclusion at the main text level.

APPENDIX E

Output-Budget and Thinking-Mode Sensitivity

This appendix labels output budget, thinking mode and continued reinforcement learning separately. The results of the same checkpoint under different output upper limits are used to measure the truncation effect; the thinking mode comparison is used to measure the impact of the explicit reasoning process; continuing reinforcement learning changes the model parameters and training process at the same time, and cannot be mixed with the static decoding comparison for interpretation. Each row in Table E1 retains its actual protocol difference.

E.1 Output Budget and Thinking Mode

Table E1a • Output budget sensitivity. Same checkpoint, same decoding mode (not thinking), only output upper limit changes; Δ relative to @2,048 lines.

Output cap	Accuracy	Δ (pp)
Stage-2C no-think @2,048	30.57%	—
Stage-2C no-think @4,096	31.35%	+0.78

APPENDIX F

State-Level Recoverability on Real Error Trajectories

This appendix answers a more rigorous question than "is it harder with longer error prefixes?": when does the student model first leave the verifiable reasoning path in real generation, whether the error state persists after leaving it, and whether the teacher can still recover in the same state. In order to answer this question, we first conduct a semantic audit on the students' real trajectory, using the starting word of the first verifiable error unit as the state anchor, thereby splitting "error entry", "error persistence" and "teacher rescue" into three observable events.

Formal sampling uses Qwen3-1.7B Base to collect 8 real trajectories for each of 200 questions on the fixed verification question pool, of which 86 questions have both correct and incorrect trajectories. The state-level part of this appendix only takes high-confidence manual review pairs of 8 of the questions, so all aggregated results are directional evidence and do not represent the overall estimate of the 200 questions.

F.1 Real Trajectories and First-Error Annotation

The state-level measurement is carried out on 8 high-confidence manual review pairs (each question contains one wrong track and one correct track for the same question), and the summary is shown in Table F2; here, one of the examples (gcd mutual prime misreading, anchor point 92 tokens) is used to fully demonstrate the reading metric definition of the same state continuation in F.3, and the remaining pairs will not be expanded card by card.

Table F1 · First verifiable error annotation of the demonstration example (measurement covers 8 review pairs, see Table F2 for summary). The anchor word offset is relative to the student's answer, excluding user prompts.

Case	Error class	First-error anchor	Failure mechanism
Case 1	Reading error · coprimality	92 tokens	gcd(N,20)=1 misread as “not divisible by 4 or 5”; factor to exclude is 2, not 4.

F.2 Observation: Can the teacher recover from the error state?

The error trace is truncated at the first verifiable error. This prefix is not added with any error correction prompts, and is handed over to the teacher model as it is, and it is asked to independently write 8 more items, and the verifier determines the final answer; the answer ratio is the teacher's recovery rate in this state. In order to eliminate the explanation that "this question is inherently difficult to answer", the correct trajectory of the same question is intercepted at the same relative position and compared with the prefix, and the same indicator is measured; the difference between the two is attributed to "being in an incorrect state" itself.

Further question: The deeper the error, the harder it is for the teacher to save it? The same error trajectory is truncated 64 words forward after the anchor point - at this time the student has written further in the error metric definition - repeat the above measurement, and compare the recovery rate at the two positions and its relative gap.

This paper calls the prefix where the starting point of continuation is taken from the wrong track as "wrong prefix", and the prefix which is taken from the same position of the correct track of the same question as "correct prefix of the same question (control)". The teacher always continues on the same prefix and does not receive additional "please correct" prompts. The results are shown in Table F2.

Table F2 · Teacher recovery rate: continuation from the wrong prefix vs continuation from the correct prefix (control) of the same question, 8 review pairs × 8 continuations.Δ = Wrong − Correct (pp); 95% CI uses prompt as the clustering unit to do bootstrap; in the root state, the two prefixes are the same text, and Δ is always 0.

State	Wrong prefix	Correct-prefix control	Δ(pp)	95% CI
Root state	21 / 64 (32.8%)	21 / 64 (32.8%)	+0.0	[+0.0, +0.0]
First-error anchor	26 / 64 (40.6%)	40 / 64 (62.5%)	-21.9	[-46.9, +0.0]
+64 tokens after anchor	12 / 64 (18.8%)	35 / 64 (54.7%)	-35.9	[-73.4, -6.3]

Table reading: When the error first occurred, the teacher had a rescue gap of about 22 pp; after going 64 words into the error trajectory, the gap expanded to about 36 pp, and the 95% CI no longer contained 0 - the damage caused by the error state to the teacher's rescue accumulated as the deviation deepened. For a case-by-case analysis of the writing process and failure locations, see F.3.

F.3 Case Study: A Complete Demonstration

The following is an example of coprime misreading (gcd, anchor point 92 words), and the F.2 reading metric definition is applied to the text: the gray box is the complete step of the student's error trajectory, the red block = the starting point of the error, the purple block = the state from the error + 64 words; for each observation position, the teacher will write two columns of the full text (red = wrong prefix, green = comparison of the correct prefix of the same question), with red words for errors and green words for corrections. The remaining review pairs are measured according to the same metric definition and are summarized in Table F2. They will not be expanded card by card.

APPENDIX F·F.3 EXAMPLE

Case 1 • gcd(N,15)=gcd(N,20)=1

Gray background = complete steps of student error trajectory (duplication removal and cleaning). Red block ① = wrong starting point; purple block ② = +64 state. There are two columns for each observation point: red = continue the full text with the wrong prefix, green = continue the full text with the correct prefix for the same question (the sample is displayed as it is, not truncated).

STUDENT ERROR TRACK · COMPLETE STEPS (CLEANING)

① Error starting point ② +64 status

```
#### Condition 3: gcd(N, 20) = 1
This means N must be coprime with 20.
- gcd(N, 20) = 1 ⇒ gcd(N, 4) = 1 and gcd(N, 5) = 1
So, N must not be divisible by 4 or 5.
#### Combining the conditions:
- gcd(N, 5) = 1 (from both 15 and 20)
- gcd(N, 4) = 1
- gcd(N, 3) = 1
So, N must be coprime with 4, 3, and 5.
This is equivalent to saying N must be coprime with 60 (since lcm(4, 3, 5)
= 60 ).
```

Position A • Error starting point t* Table F2 Anchor row

Teacher continued writing · Wrong prefix

gcd(N, 20) = 1, which means N must not be divisible by 2 **or 5**.
4. N is a 5-digit number (i.e., between 10000 and 99999), and the leading digit cannot be zero.
Let' s break this down step by step.

Step 1: Total number of 5-digit numbers with digits ⩽ 3
We are looking for 5-digit numbers where each digit is in the set 0, 1, 2, 3, but the first digit cannot be 0.
- First digit: 3 choices (1, 2, 3)
- Remaining 4 digits: 4 choices each (0, 1, 2, 3)
So the total number of such numbers is:
3 × 4^4 = 3 × 256 = 768

Step 2: Eliminate numbers divisible by 2 or 5
We are told that gcd(N, 20) = 1, so N must not be divisible by 2 or 5.
So we must exclude numbers divisible by 2 or 5.
We will use the inclusion-exclusion principle to count how many of the 768 numbers are divisible by 2 or 5.
Count of numbers divisible by 2:
A number is divisible by 2 if its last digit is even (0, 2).
- First digit: 3 choices (1, 2, 3)

- Last digit: 2 choices (0, 2)
- Middle 3 digits: 4 choices each

So total numbers divisible by 2:
$3 \times 4^3 \times 2 = 3 \times 64 \times 2 = 384$
Count of numbers divisible by 5:
A number is divisible by 5 if its last digit is 0 or 5. But we are only allowing digits $\leqslant 3$, so only 0 is allowed.
- First digit: 3 choices (1, 2, 3)
- Last digit: 1 choice (0)
- Middle 3 digits: 4 choices each

So total numbers divisible by 5:
$3 \times 4^3 \times 1 = 3 \times 64 = 192$
Count of numbers divisible by both 2 and 5 (i.e., 10):
A number divisible by 10 must end in 0 and be even (which it is, since it ends in 0).
- First digit: 3 choices (1, 2, 3)
- Last digit: 1 choice (0)
- Middle 3 digits: 4 choices each

So total numbers divisible by 10:
$3 \times 4^3 = 3 \times 64 = 192$
Use inclusion-exclusion to find numbers divisible by 2 or 5:
Divisible by 2 or 5 = 384 + 192 - 192 = 384
So, the number of numbers not divisible by 2 or 5 is:
768 - 384 = 384

Step 3: Eliminate numbers divisible by 3
We are told that gcd(N, 15) = 1, so N must not be divisible by 3 or 5. Since we already excluded numbers divisible by 5, we now only need to exclude numbers divisible by 3.
We need to count how many of the 384 numbers are divisible by 3.
A number is divisible by 3 if the sum of its digits is divisible by 3.
We are considering all 5-digit numbers with digits in 0, 1, 2, 3 and first digit not zero.
We can use the fact that the number of such numbers where the sum of digits is divisible by 3 is exactly one-third of the total (since the possible digit sums modulo 3 are evenly distributed).
So, of the 384 numbers not divisible by 2 or 5, one-third are divisible by 3.
3843 = 128
So, the number of numbers not divisible by 2, 3, or 5 is:
384 - 128 = 256

Final Answer:
256

Teacher continues writing·Correct prefix for the same question (compare)

all digits of N must be in the set 0, 1, 2, 3, and the first digit cannot be 0 (since N is a 5-digit number).
So, the first digit has 3 choices: 1, 2, 3
Each of the remaining 4 digits has 4 choices: 0, 1, 2, 3
So, the total number of 5-digit numbers with digits no greater than 3 is:
$3 \times 4^4 = 3 \times 256 = 768$

Step 2: Apply the gcd conditions
We need to find the numbers N such that:
gcd(N, 15) = 1 and gcd(N, 20) = 1

We can simplify this by noting that:
gcd(N, 15) = 1 ⇒ N is not divisible by 3 **or 5**
gcd(N, 20) = 1 ⇒ N is not divisible by 2 or 5
So, combining these, we get that N must not be divisible by 2, 3, or 5.
Thus, we need to count the number of 5-digit numbers with digits in 0, 1, 2, 3, first digit not 0, and not divisible by 2, 3, or 5.

Step 3: Use inclusion-exclusion
We count the total number of such numbers, and subtract those divisible by 2, 3, or 5.
Let’ s define the set of all such 5-digit numbers with digits ⩽ 3 and first digit ≠ 0: S = 768, as calculated above.
Let:
- A: numbers divisible by 2
- B: numbers divisible by 3
- C: numbers divisible by 5

We want to count |S (A B C)|, which is:
|S| - |A| - |B| - |C| + |A B| + |A C| + |B C| - |A B C|
We now compute each of these.

Step 4: Count numbers divisible by 2
A number is divisible by 2 if its last digit is even.
Digits allowed are 0, 1, 2, 3, so even digits are 0, 2.
So, last digit has 2 choices.
First digit: 3 choices (1, 2, 3)
Middle 3 digits: 4 choices each
So, total numbers divisible by 2:
3 × 4^3 × 2 = 3 × 64 × 2 = 384

Step 5: Count numbers divisible by 3
A number is divisible by 3 if the sum of its digits is divisible by 3.
We count how many of the 768 numbers have digit sums divisible by 3.
This is more complex, but we can use the fact that the total number of such numbers is 7683 = 256, assuming uniform distribution (which is true because digits are chosen from a uniform set and the total is divisible by 3).
So, |B| = 256

Step 6: Count numbers divisible by 5
A number is divisible by 5 if its last digit is 0 or 5.
But digits allowed are only 0, 1, 2, 3, so only 0 is allowed.
So, last digit must be 0.
First digit: 3 choices
Middle 3 digits: 4 choices each
Last digit: 1 choice (0)
So, total numbers divisible by 5:
3 × 4^3 × 1 = 3 × 64 = 192

Step 7: Count numbers divisible by 2 and 3
These are numbers divisible by 6.
We count how many of the 768 numbers are divisible by 6.
Since 6 = 2 × 3, and 2 and 3 are coprime, the numbers divisible by 6 are those divisible by both 2

and 3.
We⋯(The corpus is truncated here)

Difference: The first sentence on the side with the wrong prefix is to change 4 back to 2, continue with the correct metric definition and end with 256; the same sentence is the same on the control side, and the difference in recovery is small.

Position B · t* + 64 token table F2 +64 rows

Teacher continued writing · Wrong prefix

that:
- All digits are in 0, 1, 2, 3.
- N is **not divisible by 2, 3, 4, or 5**.
- The first digit is not 0.

So, N must be coprime with 60 (since gcd(N, 60) = 1), because lcm(15, 20) = 60.
But we can check divisibility conditions directly.

Step 2: Count all 5-digit numbers with digits ⩽ 3
We are to count all 5-digit numbers where each digit is in 0, 1, 2, 3 and the first digit is not 0.
- First digit: 1, 2, or 3 → 3 choices.
- Remaining 4 digits: 0, 1, 2, or 3 → 4 choices each.

So, total such numbers = 3 × 4^4 = 3 × 256 = 768.

Step 3: Exclude numbers divisible by 2, 3, 4, or 5
We need to count how many of these 768 numbers are not divisible by 2, 3, 4, or 5.
We use the inclusion-exclusion principle to count how many numbers are divisible by at least one of these, and subtract that from 768.
Let’ s define:
- A: numbers divisible by 2
- B: numbers divisible by 3
- C: numbers divisible by 4
- D: numbers divisible by 5

We need to find |A B C D|, and subtract it from 768.
We will calculate the number of such numbers divisible by each of the divisors, and their intersections.

Step 4: Count numbers divisible by each divisor
We will count how many of the 768 numbers are divisible by 2, 3, 4, or 5.
Divisible by 2:
A number is divisible by 2 if its last digit is even (0, 2).
- First digit: 1, 2, or 3 → 3 choices
- Last digit: 0 or 2 → 2 choices
- Middle digits: 0, 1, 2, 3 → 4 choices each

So, numbers divisible by 2:
3 × 4^3 × 2 = 3 × 64 × 2 = 384
Divisible by 3:
A number is divisible by 3 if the sum of its digits is divisible by 3.
We can count this by checking all 768 numbers and counting how many have digit sums divisible by 3.
But it's easier to compute the number of such numbers using modular arithmetic.
Each digit is 0, 1, 2, or 3. So, the total number of 5-digit numbers with digits ⩽ 3 is 768.

We can use the fact that the number of such numbers with digit sum divisible by 3 is approximately 13 of the total, because the digits are evenly distributed modulo 3.
So, numbers divisible by 3: 7683 = 256
Divisible by 4:
A number is divisible by 4 if its last two digits form a number divisible by 4.
We can count this by considering all possible last two digits (from 00 to 33) that are divisible by 4, and for each, count the number of combinations for the first 3 digits.
There are 16 possible two-digit combinations (from 00 to 33), and only 4 of them are divisible by 4: 00, 04, 08, 12, 16, 20, 24, 28, 32, 36 → but only those ⩽ 33 are valid.
Valid two-digit combinations divisible by 4: 00, 04, 08, 12, 16, 20, 24, 28, 32 → 9 combinations.
So, numbers divisible by 4:
3 × 4^2 × 9 = 3 × 16 × 9 = 432
Divisible by 5:
A number is divisible by 5 if its last digit is 0 or 5. Since digits can only be 0⋯(The corpus is truncated here)

Teacher continues writing·Correct prefix for the same question (compare)

means N must not be divisible by any of the prime factors of 15, which are 3 and 5.
So, N must not be divisible by 3 **or 5**.
Condition 3: gcd(N, 20) = 1
This means N must not be divisible by any of the prime factors of 20, which are 2 and 5.
So, N must not be divisible by 2 or 5.
Combining conditions 2 and 3, N must not be divisible by 2, 3, or 5.
So, the number N must be coprime with 30 (since lcm(15, 20) = 60, and the gcd condition implies gcd(N, 30) = 1).
Thus, we are looking for 5-digit numbers N with digits in 0, 1, 2, 3, first digit not zero, and gcd(N, 30) = 1.

Step 2: Count all 5-digit numbers with digits in 0, 1, 2, 3, first digit not zero
We count the number of such numbers:
- The first digit: 3 choices (1, 2, 3)
- The remaining 4 digits: 4 choices each (0, 1, 2, 3)
So, total such numbers = 3 × 4^4 = 3 × 256 = 768

Step 3: Remove numbers divisible by 2, 3, or 5
We need to count how many of these 768 numbers are not divisible by 2, 3, or 5.
This is a classic inclusion-exclusion problem.
Let’ s define:
- A: numbers divisible by 2
- B: numbers divisible by 3
- C: numbers divisible by 5
We want to count the numbers not in A B C, i.e., numbers not divisible by 2, 3, or 5.
So we compute:
Valid numbers = Total numbers - |A B C|
Using inclusion-exclusion:
|A B C| = |A| + |B| + |C| - |A B| - |A C| - |B C| + |A B C|
We now compute each term.

Step 4: Count numbers divisible by 2
A number is divisible by 2 if its last digit is even: 0, 2

So, among the 4 choices for each digit (0, 1, 2, 3), 2 are even.
Thus, the number of 5-digit numbers with digits in 0, 1, 2, 3, first digit not zero, and last digit even is:
- First digit: 3 choices
- 3 middle digits: 4 choices each
- Last digit: 2 choices (0, 2)
So, |A| = 3 × 4^3 × 2 = 3 × 64 × 2 = 384

Step 5: Count numbers divisible by 3
A number is divisible by 3 if the sum of its digits is divisible by 3.
Let’ s count how many of the 768 numbers have digit sums divisible by 3.
This is more complex, but we can use modular arithmetic.
Each digit can be 0, 1, 2, or 3. Let’ s denote the digits as d_1, d_2, d_3, d_4, d_5, where d_1 1, 2, 3, and the rest d_i 0, 1, 2, 3.
Let’ s compute the total number of such 5-digit numbers with digit sum divisible by 3.
This is a standard problem in combinatorics. The total number of such numbers is:
13 × Total numbers = 13 × 768 = 256
So, |B| = 256

Step 6: Count numbers divisible by 5
A number is divisible by 5 if its last digit is 0 or 5.
But our⋯(The corpus is truncated here)

Difference and interpretation: After delving into 64 tokens, the constraint on the wrong prefix side is written as "2, 3, 4, and 5 are all excluded" and is called gcd(N,60)=1; the control side is still excluding only 2, 3, and 5. Mistakes will lead to bias in the follow-up, and the teacher cannot save them.

F.4 Evidence Chain and Interpretation Boundaries

The chain of evidence unfolds over four events. Among the 200 questions, 8 questions each, and a total of 1,600 real trajectories, 1,285 failed the result verification, and 86 questions had both correct and incorrect trajectories, indicating that the state accessed by the model did not unfold along a single correct path, but contained comparable success and failure branches. Before entering state-level analysis, only the trajectory that can locate the first verifiable semantic error is retained; repeated formats, answer extraction failures, and truncation are excluded separately to avoid misjudgment of output protocol issues as inference deviations. Then, the wrong trajectory and the correct trajectory are matched on the same question and the same relative position; only when the teacher's rescue rate in the error state is lower than the control, the difference can be attributed to the state conditions rather than the difficulty of the question. Finally, the status of the 64 tokens after the anchor point is used to test whether the error continues to constrain subsequent production: the error-control gap of teacher rescue in the pilot expanded with deviation, supporting the directional explanation that "error states are cumulative".

The implication for online distillation is that OPD imposes local token supervision on the state that the student has visited. Once the state deviates from the verifiable path, the teacher's local distribution may not still be a reliable error correction target. Therefore, the quality of student status access is a key moderator variable of OPD health, but it is not the only factor.

It should be emphasized that the teacher's continued writing is not a forced rewrite of the student's answer, but "continues to generate the student's prefix" and then scores based on the final answer - it does not have any ability to modify the text that the student has written. Therefore, once the student's trajectory enters the wrong solution space, the prefix obtained by the teacher is itself a continuation of the wrong derivation, and its error correction ability will also become invalid; this is the mechanism explanation of the rescue rate decaying with +64 words in Table F2, and is also consistent with the "distillation is only applied when the teacher is trustworthy" in 4.3 of the main text.

The strongest statement supported by this appendix is that error states in students' true production are related to the teacher's subsequent recoverability, and that the difference increases as deviations persist. It does not support "the teacher cannot recover all error states", nor does it support the global collapse of the model using entropy alone. This paper uses "trajectory-level irrecoverability" separately from entropy collapse and reward collapse during the training process; the latter must be separately proven by step-by-step training dynamics.

F.5 Data and Reproduction Artifacts

All data and artifacts are stored in the project artifact directory: 200 student real tracks, 86 same-question paired audit queues, 8 manual review records, teachers' same-status continuation samples, and summary JSON and chart source scripts. Formal expansion requires increasing the scale of manual review to at least 20–30 question pairs (prompt-level pairings) and re-estimating the overall rescue difference accordingly.

APPENDIX G

Robustness under Multiple Sampling (Pass@k)

This appendix is a supplementary robustness check for the single sampling results of fixed 1,024 questions. It is used to confirm that the gating method does not introduce additional diversity loss; it is descriptive evidence and does not participate in the significance test.

Fixed verification uses a single sampling and cannot distinguish between "stable mastery of the model" and "occasionally correct answers". In order to characterize the multiple sampling coverage capabilities of each method, this paper samples each question 8 times with temperature 1.0 on a 128-question subset of the fixed 1,024-question validation set (the protocol is consistent with the main evaluation), and reports the cumulative proportion of correct answers at least once Pass@\(k\) (Figure G1). GRPO is not inferior to other methods in \(k\ge2\), and the curves of AGOPD and RL+OPD are close, indicating that gating also does not introduce additional diversity loss under the multi-sampling perspective, consistent with proposition 3. This subset is intercepted in file order and the difficulty is on the easy side. The gap between methods is compressed by the ceiling effect. This figure is only used as descriptive robustness evidence and does not participate in significance testing.

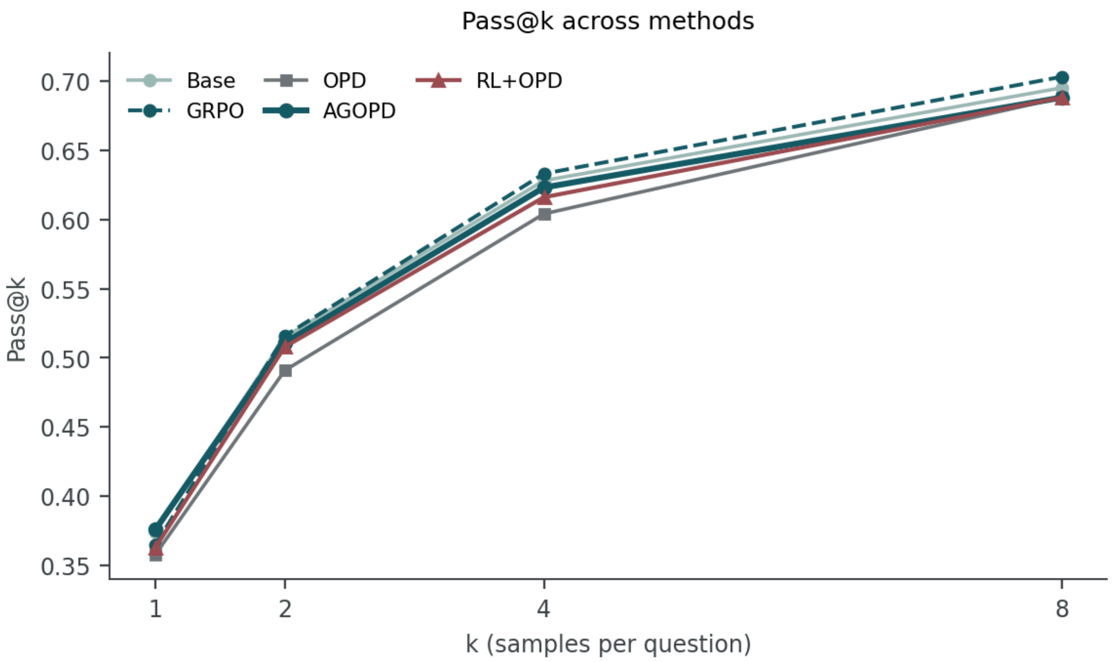


Figure G1 • Multiple sampling coverage (Pass@k).

APPENDIX H

Backbone Scale and Signal Density: Easier Exploration as a Candidate Solution

Chapters 3 and 4 posit the negative transfer of unconditional online distillation as a state dislocation problem: students with small bases are frequently in wrong states, and teachers lose part of their recoverability in these states. This appendix presents solution evidence in the same direction along the scale axis: a stronger base makes it easier to access the correct exploration state under the same sampling protocol, which both directly improves the actual benefit of the resultant supervision RL and may also dilute the exposure of state misalignments that impair OPD. The following lists the census signal density (Table H1), the actual effect of monitoring RL with the same budget results (Table H2), and finally explains the connection between this scale of evidence and the previous mechanism analysis and the boundaries to be verified.

H.1 Census protocol and signal density

The census uses the same clean chat rendering and rule scoring as A.1: 1.7B and 4B both use clean-chat with the same protocol as Appendix A.1. Full census: $n=8$ non-thinking decoding of the same 16,164-question training set, question-by-question, and scoring by the rule validator. Taking the size of the base as the axis, the three indicators move in the same direction: the proportion of questions with all eight sampling errors decreased, the average success rate of questions increased, and the signal density that can form a contrast between positive and negative groups under a fixed group size increased (Table H1). The rankings are consistent when the group sizes are 4, 6, and 8, indicating that this conclusion does not depend on a single group size (Table H1).

Table H1 • Census signal density as a function of base size. S(G) = $\mathbb E_{\hat p}[1-(1-\hat p)^G-\hat p^G]$; 1.7B and 4B are both the same 16,164-question clean-chat full census (same protocol and scoring standards as Appendix A.1), and can be directly compared based on base size.

Base model	Census n	$\hat p=0$ (95% CI)	mean $\hat p$	S(4)	S(6)	S(8)
1.7B (full)	16,164	48.9% [48.1, 49.7]	0.232	0.301	0.360	0.394
4B (full)	16,164	39.4% [38.6, 40.1]	0.309	0.344	0.411	0.449

H.2 Actual effect: Outcome-supervised RL under matched budgets

Signal density is just a quantity of data conditioning; whether the base size is truly converted into results depends on the resultant supervised RL output under the same training budget. Under the fixed 1,024 questions, non-thinking@2,048, and seed 42 outcome accuracy rate (reward metric definition, rule validator judgment), the 50-step GRPO is significantly higher than 1.7B on 4B (Table H2). The comparison has the same 50 steps and the same fixed evaluation metric definition, but the details of the historical training batches and update channels of the two runs are not completely controlled, and the difference is used as a directional reference. What is more noteworthy is that the starting point of 4B itself is higher, and the GRPO increment relative to the same budget of the respective Base is also larger (Table H2): a stronger base not only provides a higher a priori starting point, but also can make fuller use of the outcome supervision signal of the same budget. At the training process level, the larger base also enters faster and maintains higher online outcome scores under the same lr=1e-6 full parameter outcome supervision GRPO protocol (Figure H1), which is in the same direction as the fixed evaluation difference.

Table H2 • Results The actual effect of supervised RL varies with the size of the base (reward metric definition: the outcome accuracy of the rule validator; fixed 1,024, non-thinking@2,048, seed 42). This metric definition is consistent with the Base value of the fixed evaluation in the text; both scales of GRPO are 50 steps, and Δ is relative to their respective Bases.

Base model	Base (reward acc., fixed 1,024)	GRPO-50 (reward acc.)	Δ (pp; GRPO - Base)
1.7B	23.3%	25.4%	+2.1 pp
4B	30.2%	40.2%	+10.0 pp

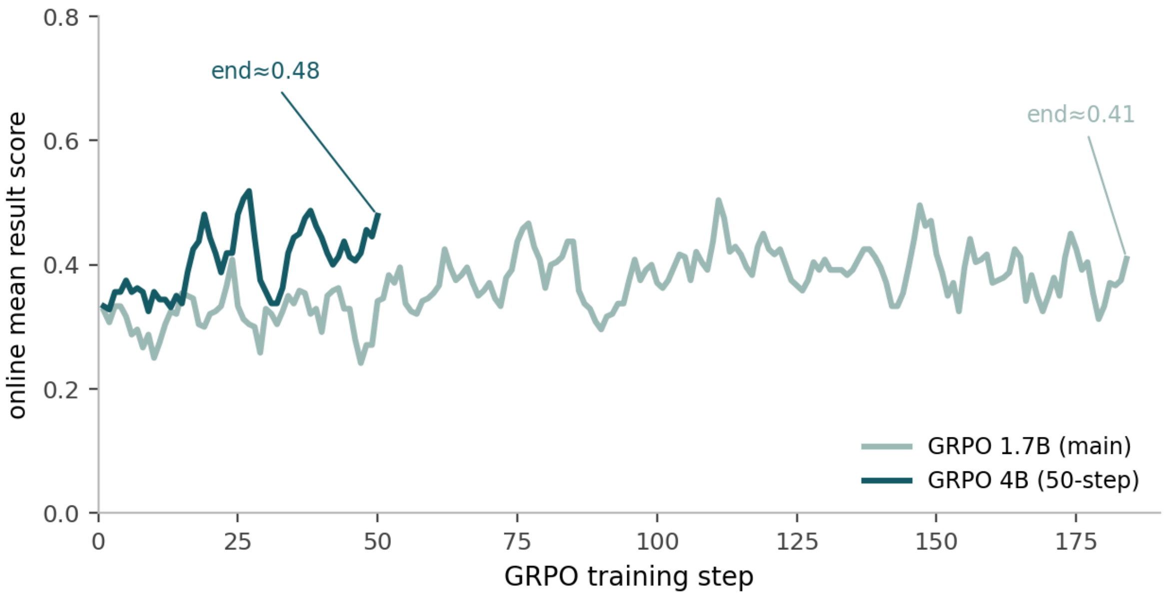


Figure H1 • Result of supervised GRPO training curve with base scale. The online average outcome score (rule score) changes with optimization steps, 5-step moving average: 4B (50 steps) enters and maintains a higher range faster, 1.7B main baseline is overall low and arrives slower. Both runs are lr=1e-6 fully parameterized outcome-supervised GRPO; the x-axis is the respective optimization step, and the non-synchronous time is aligned.

H.3 From Mechanistic Analysis to a Candidate Solution

The previous mechanism analysis gives the direct source of negative transfer: small-base students visit frequently on the wrong path, and the teacher's recoverability in these states decreases as the errors deepen (§ 3.3). Scale evidence responds to this source from both ends. First, a stronger base reduces the proportion of completely wrong questions, increases the average success rate, and increases the signal density under a fixed group size (Table H1), indicating that RL groups more often contain positive samples and learnable positive and negative contrasts are easier to form. Second, the actual benefits of result-supervised RL increase with scale under the same training budget (Table H2), indicating that a stronger exploration space translates into final results. Putting these two points together, this paper considers scale improvement as a candidate solution to state dislocation-type negative transfer: it reduces the dependence on teacher rescue and dilutes the deep fault state exposure that damages OPD.

The boundary of this direction needs to be explicitly stated: first, the OPD/gated four-arm controlled control of 4B students has not yet been run, and this appendix cannot directly claim that scale eliminates negative transfer, and only gives two pieces of evidence in the same direction, signal density and actual RL effect; second, the batch and channel details of the two runs are not completely controlled, and the actual effect difference is a directional reference; third, teacher selection (question-level advantage) is orthogonal to scale improvement, and scale evidence does not replace the gating mechanism in §4. The previous chapters are mechanism analyses, and the scale paths given in this appendix are solution directions that need to be confirmed by controlled experiments.

APPENDIX I

The Entropy–Signal Dilemma in On-Policy Distillation: Anchors and the Limits of KL/Entropy Penalties (Exploratory)

Focusing on "how to learn the teacher's correct signal through online token-level distillation", in the default configuration, hard-CE distillation first occurs position by position on the entire student context (the target is the teacher argmax); this paper then tries to tighten its position and only apply it at the disagreement between the teacher and the student or the teacher's confidence, in order to leave the signal and remove the uninformative component. This tightening will also remove the implicit effect of the "overlap self-distillation anchor" in the main configuration of AGOPD, making the policy entropy unconstrained; increasing KL intensity or adding an entropy penalty can only postpone the entropy runaway, but cannot cure it. This appendix records the results of this batch of variant experiments: all variants did not exceed the pure outcome supervision baseline. Based on this, "whether online token-level distillation can bring gains" is narrowed down to a mechanical boundary rather than a hyperparameter issue.

I.1 The role of anchors

In the default main configuration, hard-CE distillation first occurs position-by-position over the entire student context: the target is the teacher's argmax at that position. The student has correctly overlapped the position, and the target is consistent with the token that the student has generated. Hard-CE here degenerates into "following oneself", which is equivalent to embedding the self-distillation entropy anchor into the distillation itself; it does not transmit new signals, but anchors the strategy near itself, maintaining observable entropy and learning stability. The cost is that the anchor is coupled to the signal, and it is impossible to only retain the "informative" distillation - narrow the distillation to the position where the teacher argmax is different from the student sample (different from pure OPD, different from AGOPD), or further require the teacher to be double-confident (teachable) at this position, which will also remove the anchor.

I.2 Soft target to hard target: entropy channel

Before switching to hard targets, AGOPD used teacher top-16 distribution and soft targets sharpened by temperature 0.7 for distillation supervision. Figure I1 compares the policy entropy trajectories of the two versions under the same comfort pool and two-level gating: the hard (teacher argmax) is low and stable throughout; the soft target version is overall higher and drifts upward with each step, and does not return to the low entropy channel of the hard version before premature termination at step 94. This observation is the direct motivation for changing the distillation target to the teacher argmax. It needs to be distinguished from the anchor deletion variant (Figure I2) described later: the entropy drift of the soft version is limited and does not enter the policy entropy out-of-control zone; complete loss of control only occurs in divergence variants with more complete anchor deletion.

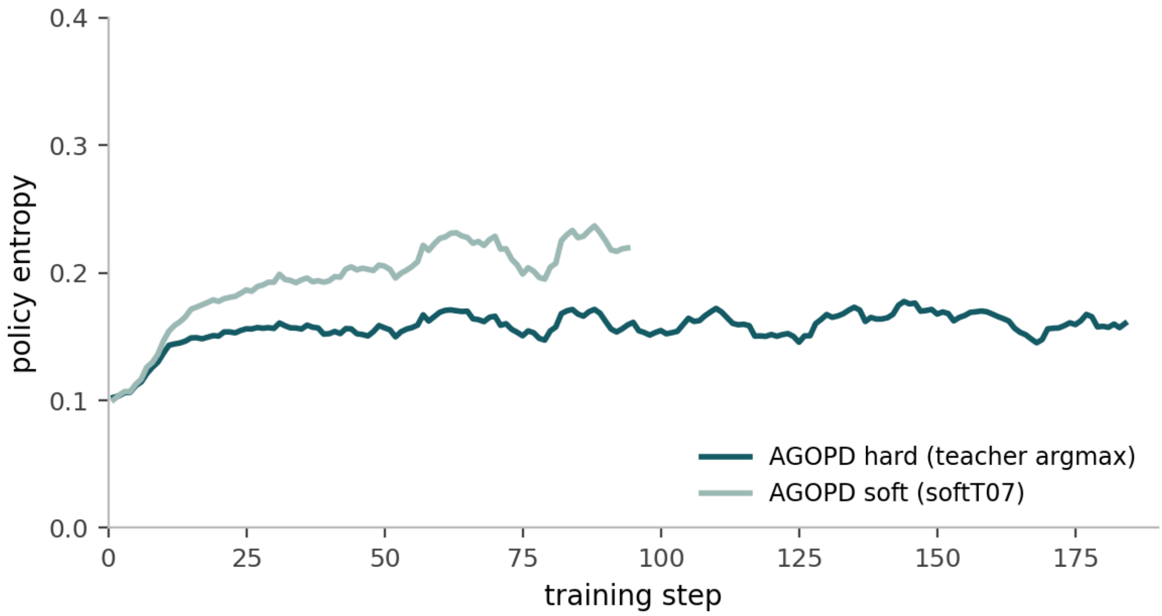


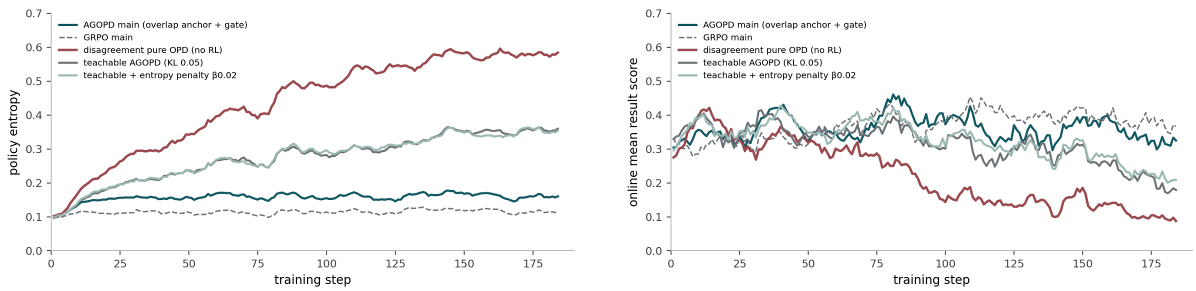
Figure I1 · AGOPD distillation target: policy entropy trajectory for soft (softT07) versus hard (teacher argmax). The hard target is low and smooth throughout, the soft target is higher overall and drifts upward. The two versions have the same pool (comfort) and the same two-level gating, with only different distillation targets; the soft version terminates early at step 94. The curve is a 9-step moving average.

I.3 Results

The observations of variant training (Table I1, Figure I2) are consistent: once the overlap anchor is deleted, the policy entropy rises step by step and eventually goes out of control, and the online result points fall simultaneously, with the pure OPD being the most serious difference; the teachable variant and the entropy penalty variant can only postpone the loss of control, and the entropy penalty also drags down the final score; increasing the KL coefficient also only delays the loss of control. The anchor-preserving AGOPD main configuration and the pure outcome supervision baseline are used as healthy controls, with stable entropy and stable outcome scores.

Table I1 • Online distillation entropy—signal dilemma variant. Online entropy is the stepwise actor entropy (first value → last value or peak); fixed 1,024 is the semantic accuracy rate (see § 5.1 for GRPO/AGOPD lines). All variants did not exceed the GRPO baseline of pure outcome supervision.

Variant	Online entropy path	Fixed 1,024	Note
AGOPD main (overlap anchor + gates)	peak ⩽0.23	27.05%	reference (§ 4/ § 5)
GRPO main (no distillation)	⩽0.17	28.42%	result-only baseline
disagreement pure OPD (no RL)	0.10 → 0.62	8.01%	no RL; hard-CE only at disagreement
teachable AGOPD, KL 0.05	→ 0.38	19.04%	dual-confidence gate
teachable + entropy penalty β0.02	→ 0.39	17.48%	entropy-penalty variant
disagreement AGOPD, KL 0.05	→ 0.40 @ step 94	—	guard-killed at step 94
disagreement AGOPD, KL 0.2	→ 0.23 @ step 22	—	interrupted at step 22



(a) Policy entropy changes with training steps

(b) Online results are divided into training steps

Figure I2 • Training dynamics of the online distillation variant (entropy-signal dilemma).(a) The entropy of the variant strategy that deletes the overlap anchor continues to climb, and the divergence of pure OPD is the most serious; (b) its online outcome score collapses. The main configuration of the anchor with the GRPO baseline was retained as a healthy control. The curve is a 9-step moving average.

I.4 Convergence Implications

This batch of explorations is consistent with the convergent implications of the text. The "entropy constraint" and "signal selectivity" of online distillation cannot be decoupled: distilling only where there is a signal will also remove the implicit entropy anchor. External brakes such as KL and entropy penalty cannot replace structural anchoring. Therefore, AGOPD does not adopt the "purer" variant at the word level, but retains the overlap anchor and places the selectivity in the outcome-level gate (advantage gate and teacher competency gate). And because all variants do not exceed the pure outcome supervision baseline, online token-level distillation is reduced to a mechanical conclusion of no gain under the settings of this paper, and distillation only occurs within the scope limited by the result gate.

Inference-Engine and Rollout-Topology Optimization (Engineering Appendix)

All the online training and state-level probe evidence in this paper are undertaken by the rollout engine composed of verl v0.8 (FSDP training loop) and vLLM 0.10 (sampling and teacher inference). The throughput of multi-GPU rollout is not only determined by a single parameter of the engine, but also by the topology of the training process and sampling process (process distribution, parallel segmentation, weight synchronization method) as a whole. This appendix records the configuration optimization carried out to make online token-level distillation on 4×A100 feasible in the form of engineering reproduction instructions: first, the baseline topology and observed bottlenecks are given, and then the configuration changes and measured differences are given. This description is an engineering reproduction attachment and does not constitute a scientific comparison at the method level.

J.1 Baseline: Process-separated standalone topology and bottleneck observation

The initial deployment places the training actor, student sampling and teacher inference as three independent processes on separate GPUs: the student uses a single FSDP card to handle the optimization step, a separate vLLM server handles student rollout, and another separate vLLM server handles teacher inference, leaving the remaining GPUs idle. The inference engine adopts conservative parameters: turning off CUDA Graph (`enforce_eager=True`), low memory quota, batch processing limit of 16K tokens, and default number of concurrent workers. The precise values of each parameter are shown in Table J1. After each training step, the student weights need to be broadcast to the sampling server across processes, which constitutes a fixed overhead.

Bottleneck observations can be summarized into three points. First, in the composition of the total time-consuming of a single step, in addition to generation, reference forward, actor update and cross-process weight synchronization all account for a significant proportion. There are a large number of serial waits between stages, and the GPU idle window is obvious. Second, cross-process weight synchronization takes a fixed time of ten seconds per step, regardless of the sampling volume. Third, increasing the vLLM GPU memory quota did not bring about changes in end-to-end throughput, indicating that the generation under this configuration is not constrained by the KV cache capacity; it is a misjudgment to treat the spare GPU memory as a target that must be filled. The time consumption of the stages corresponding to the above observations is shown in Figure J1.

J.2 Optimization: Joint changes of sampling topology and engine parameters

The optimization follows the principle of "decomposing bottlenecks by end-to-end step time consumption and attributing changes by stage". The changes are divided into two layers: the topology layer reuses verl's actor/rollout hybrid deployment (`rollout.nnodes=0`), so that the sampling replica and the actor coexist on the same card, canceling the independent rollout process and the weight broadcast it brings; the parallel layer changes the student to 2-way FSDP, while the teacher keeps a single-GPU TP1, single-replica inference process and does not use TP2. The rationale is that for long generations (2,048 tokens) throughput is bounded by the number of concurrent requests rather than single-request latency, whereas tensor parallelism requires per-layer communication and buys little on latency; should teacher-side concurrency need to grow, the intended extension is TP1 with multiple replicas (data parallel) rather than TP2 - this extension was not enabled in the runs reported here. A single GPU suffices to host the teacher: 4B at TP1, and the 8B teacher occupies about 72 GB at a 0.85 memory fraction.

The engine layer simultaneously enables vLLM sleep mode and gradual KV cache release: the engine releases weights in the generation gap and restores them on demand. Together with only synchronizing separable deltas (LoRA delta semantics), cross-process weights are synchronously compressed to sub-second levels. On the concurrent side, increase the number of AgentLoop workers and test a higher batch processing limit (Table J1). CUDA Graph (eager turned off) is available at the single-engine level, but it cannot enter the first training step within the limited time under this multi-process combination, and there is interaction between sleep/resume and graph capture, so its activation is restricted to "first pass end-to-end smoke verification before entering the main experiment."The GPU memory quota is changed to only increase when a KV insufficient alarm occurs. The configuration comparison before and after the change is shown in Table J1.

Table J1 • Sampling—Comparison of inference topology and minimum configuration before and after engine parameter optimization. For detailed design reasons and boundary conditions, see the main text of J.2.

Aspect	Baseline (standalone)	Optimized (hybrid)
Topology	separate processes on separate GPUs	actor/rollout co-located (`rollout.nnodes=0`)
Student	FSDP, 1 GPU	FSDP 2-way
Rollout	1 standalone server	TP1 hybrid replicas
Teacher	1 standalone server (TP1)	single-GPU TP1, 1 replica (no TP2)
Weights sync	full-weight broadcast	sleep-mode delta (<1 s)
Workers / max tokens	8 / 16,384	16 / up to 32,768
VRAM / CUDA graph	gmu 0.25 / eager	raise only on KV pressure / graph gated on e2e

J.3 Measured differences

The end-to-end difference is recorded with the same 1.7B GRPO+OPD (E3 category) task and the same sampling budget. The speedup brought by the hybrid topology does not come from the generation itself, but from the reference/scoring, actor update and weight synchronization stages; the phased comparison and the end-to-end total time consumption are shown in Figure J1; the precise values, metric definitions and disclaimers are given with the legend.

−27%

End-to-end single step total time consumption chart J1

−33%

Reference and Reward/Grading Stage Figure J1

−32%

Actor update (2-way FSDP) Figure J1

−35%

Weight synchronization (single-step file) Figure J1 Annotation metric definition

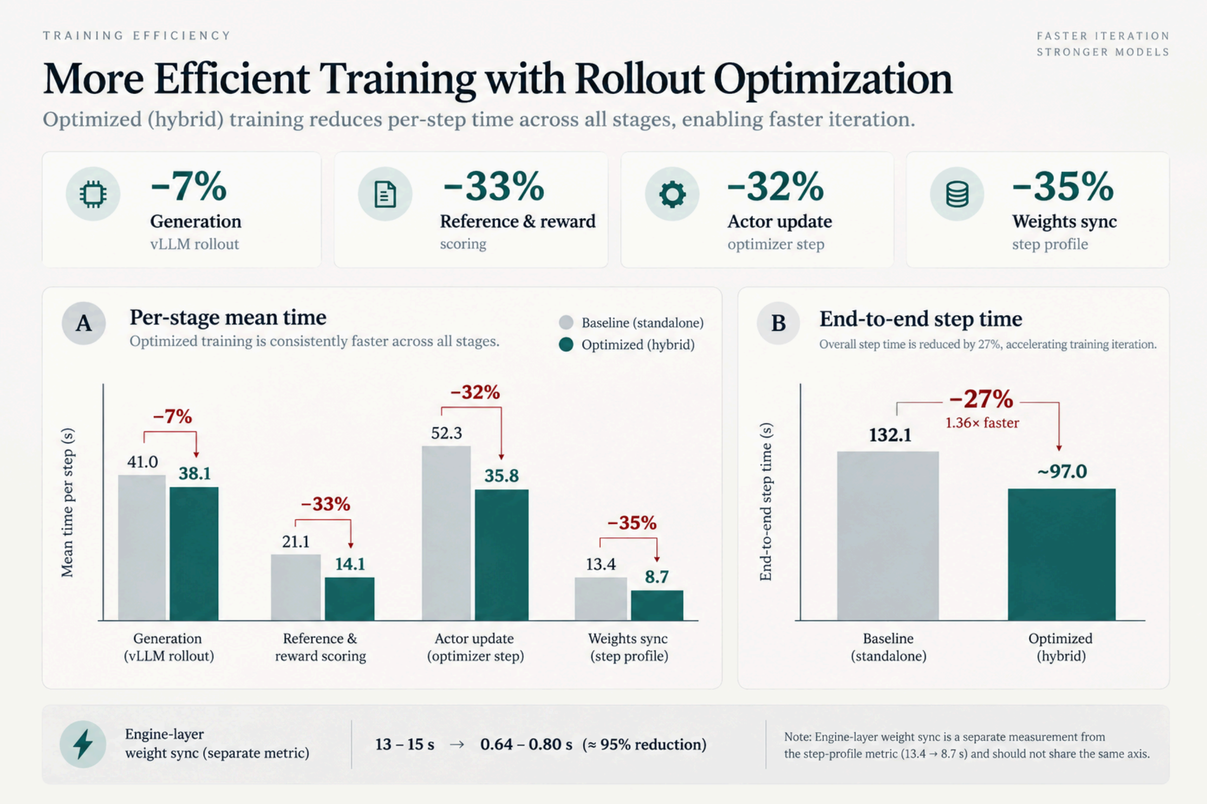


Figure J1 • Comparison of rollout topology optimization before and after. The two levels of gray/main color correspond to standalone before optimization and hybrid after optimization respectively. The reduction in stage and total time consumption is marked as a percentage. The data comes from adjacent task records of the same Qwen3-1.7B GRPO+OPD (E3 category) task with a 2,048 word budget: the sum of each stage is not equal to the end-to-end step total (the step total also includes unbroken overhead such as data assembly, scheduling waiting, etc.). The actual weight synchronization (sleep-mode) of the engine layer is 0.64–0.80 s, which is different from the step file diameter in the figure; the time-consuming comparison of adjacent task instances is not used as an independent controlled benchmark.

Attribute the benefits to specific changes: actor updates are significantly shortened due to 2-way FSDP; topology convergence reduces reference/scoring and scheduling waits; sleep mode incremental synchronization reduces cross-process weight synchronization to sub-second level. The limited changes in the generation phase indicate that generation is not the main bottleneck in this workload.

J.4 Experience and Generalizable Rules

- The throughput evaluation should be based on the end-to-end step time and its stage decomposition. The average GPU utilization does not constitute an equivalent indicator: the utilization does not increase as a whole as each stage alternates and waits.
- The GPU memory quota is only increased when insufficient KV capacity occurs; free GPU memory is not equal to the resource target that must be filled.

- Any engine parameter changes (CUDA Graph, batch processing limit, worker concurrency) must be verified through end-to-end smoke verification before entering the main experiment. Single engine layer availability cannot be extrapolated to multi-engine combinations.
- Weight synchronization and process distribution are low-hanging fixed overheads that take precedence over single-point parameter adjustment of the generation engine; this optimization ultimately supports the main experiment and each probe to run in a 3+1 hybrid topology (three-card actor/rollout multiplexing, one-card teacher).

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This version deliberately distinguishes source-supported results, post-hoc analysis, and unresolved limitations. Values without recoverable artifacts are excluded from core quantitative claims.